

Additional Practice

ACC7.1.01

1. Determine whether each transformation shows a translation, reflection, or rotation.

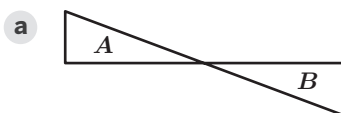


Translation

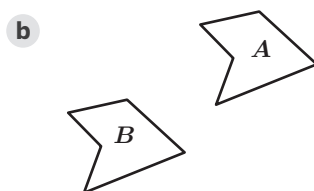


Rotation or reflection

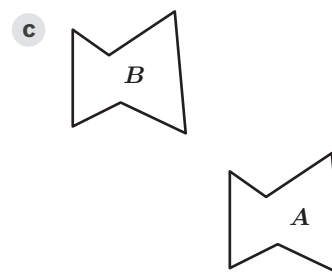
2. Determine whether the transformation from figure *A* to figure *B* shows a translation, reflection, or rotation.



Rotation



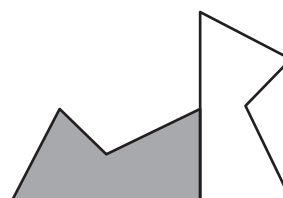
Translation



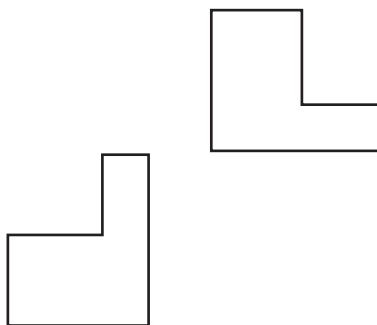
Translation

3. Describe the transformation.

Rotation



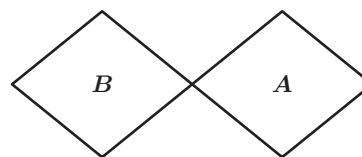
4. Describe the transformation.



A rotation OR a rotation and a translation

5. Which transformation could move figure *A* to figure *B*?
Select *all* that apply.

- ☒ A. Reflection
- ☒ B. Rotation
- ☒ C. Translation
- ☐ D. None of these

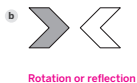


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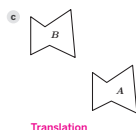
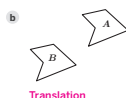
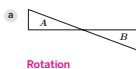
Additional Practice

ACC7.1.01

1. Determine whether each transformation shows a translation, reflection, or rotation.



2. Determine whether the transformation from figure A to figure B shows a translation, reflection, or rotation.



3. Describe the transformation.

Rotation



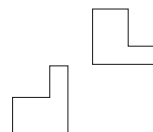
Unit 1 Lesson 1

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Name: _____ Date: _____ Period: _____

4. Describe the transformation.



A rotation OR a rotation and a translation

5. Which transformation could move figure A to figure B? Select *all* that apply.

- ☒ A. Reflection
☒ B. Rotation
☒ C. Translation
☐ D. None of these



Unit 1 Lesson 1

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.G.A.1
2	1	8.G.A.1
3	1	8.G.A.1
4	1	8.G.A.1
5	2	8.G.A.1

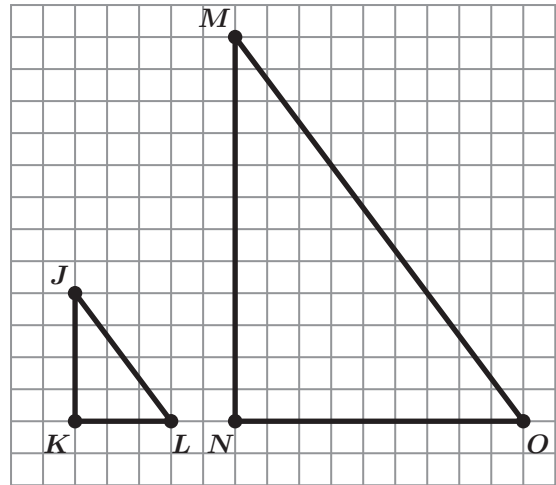
Notes:

Additional Practice

ACC7.2.01

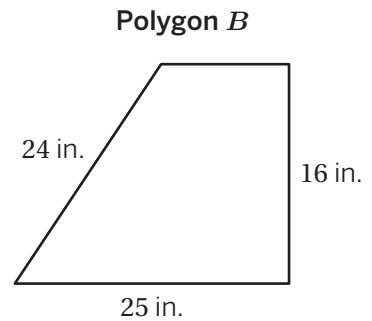
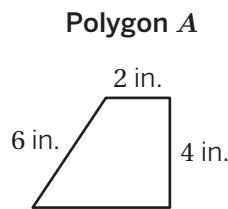
1. Triangle MNO is a scaled copy of triangle JKL . What is the scale factor that takes triangle JKL to triangle MNO ?

- A. 2
- B. 3**
- C. 4
- D. 5



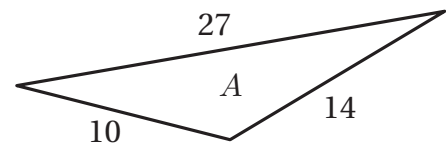
2. Polygon B is a scaled copy of polygon A .

- a What is the length of the shortest side of polygon B ?
8 inches
- b What is the length of the longest side of polygon A ?
6.25 inches (or equivalent)



3. Figure A is scaled by a factor of $\frac{1}{2}$. Select *all* three side lengths of the scaled copy.

- ☒ A. 7
- ☒ C. 13.5
- ☒ E. 5
- ☐ B. 28
- ☐ D. 20
- ☐ F. 54

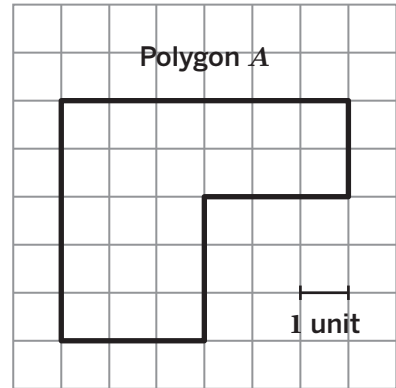


4. Polygon A has side lengths of 10, 18, 20, and 22 units. Polygon B is a scaled copy of polygon A . Which of the following sets of values could *not* be the side lengths of polygon B ?

- A. 5, 9, 10, 11 B. 25, 45, 50, 55
C. 20, 28, 30, 32 D. 2, 3.6, 4, 4.4

5. Polygon B is a scaled copy of polygon A . The shortest side of polygon B is 13.5 units.

- a What is the scale factor that takes polygon A to polygon B ?
6.75 (or equivalent)
- b What is the length of the longest side of polygon B ?
40.5 (or equivalent)



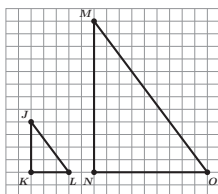
Name: _____ Date: _____ Period: _____

Additional Practice

ACC7.2.01

1. Triangle MNO is a scaled copy of triangle JKL . What is the scale factor that takes triangle JKL to triangle MNO ?

A. 2
B. 3
 C. 4
 D. 5



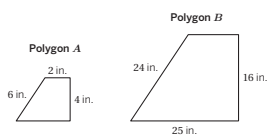
2. Polygon B is a scaled copy of polygon A .

a. What is the length of the shortest side of polygon B ?

8 inches

b. What is the length of the longest side of polygon A ?

6.25 inches (or equivalent)



3. Figure A is scaled by a factor of $\frac{1}{2}$. Select all three side lengths of the scaled copy.

☒ A. 7

☐ B. 28

☒ C. 13.5

☐ D. 20

☒ E. 5

☐ F. 54



Unit 2 Lesson 1

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Name: _____ Date: _____ Period: _____

4. Polygon A has side lengths of 10, 18, 20, and 22 units. Polygon B is a scaled copy of polygon A . Which of the following sets of values could not be the side lengths of polygon B ?

A. 5, 9, 10, 11
C. 20, 28, 30, 32

B. 25, 45, 50, 55
 D. 2, 3, 6, 4, 4, 4

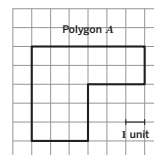
5. Polygon B is a scaled copy of polygon A . The shortest side of polygon B is 13.5 units.

a. What is the scale factor that takes polygon A to polygon B ?

6.75 (or equivalent)

b. What is the length of the longest side of polygon B ?

40.5 (or equivalent)



Unit 2 Lesson 1

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	7.G.A.1
2	2	7.G.A.1
3	2	7.G.A.1
4	2	7.G.A.1
5	2	7.G.A.1

Notes:

Additional Practice

1.10

Problems 1–4: James and Piper buried a treasure together on their school's field. The field is 400 feet wide. James made a map that is 8 inches wide to record its location.

1. Write two possible scales James could have used to make his drawing.

Explanations vary. James could use a scale that every 1 unit is 4 inches wide. James could also use a scale that every 2 units is 8 inches wide.

2. Piper made her own map using a scale of 1 inch to 20 feet. Whose map is larger: James's or Piper's? Explain your thinking.

Explanations vary. Since James is possibly using a scale of 4 inches for every unit, this is smaller than Piper's scale. So, James' map will need to be larger.

3. On Piper's map, the treasure is 2 inches from the south edge of the field. How far is the treasure from the south edge on Piper's map?

Since Piper is using a scale of 1 inch to 20 feet, the treasure is 40 feet from the south edge of Piper's map.

4. On Piper's map, the area of the field is 16 square inches. Piper says that the actual area of the field is 320 square feet. Is Piper correct? Explain your thinking.

Explanations vary. Piper is incorrect. Piper did not correctly scale the length and width of the field first. She needed to find the actual area of the length and width first. Then, multiply to find the area.

5. Select *all* the scales that are equivalent to 4 centimeters to 20 meters.

- ☐ A. 4 inches to 20 inches
- ☒ B. 1 centimeter to 5 meters
- ☒ C. 10 meters to 2 centimeters
- ☐ D. 5 millimeters to 3 meters
- ☐ E. 1 inch to 4 feet

Explain your thinking for the scale(s) you selected.

Explanations vary. Since the scale is 4 centimeters to 20 meters, I can divide both sides by 4 to get an equivalent scale of 1 centimeter to 5 meters. I can also divide both sides of the scale by 2 and flip the order, to get the equivalent scale 10 meters to 2 centimeters.

6. On a blueprint, the living room is 4.2 inches wide. The blueprint has a scale of 1 inch to 5 feet. How wide would the living room be on a blueprint that has a scale of 1 inch to 20 feet?

- A. 1.2 inches
- ☒ B. 2.1 inches
- C. 3.2 inches
- D. 31 inches

Name: _____ Date: _____ Period: _____

Additional Practice

1.10

Problems 1–4: James and Piper buried a treasure together on their school's field. The field is 400 feet wide. James made a map that is 8 inches wide to record its location.

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Explanations vary. James could use a scale that every 1 unit is 4 inches wide. James could also use a scale that every 2 units is 8 inches wide.

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Explanations vary. Since James is possibly using a scale of 4 inches for every unit, this is smaller than Piper's scale. So, James' map will need to be larger.

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4. On Piper's map, the area of the field is 16 square inches. Piper says that the actual area of the field is 320 square feet. Is Piper correct? Explain your thinking.

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Unit 1 Lesson 10

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Name: _____ Date: _____ Period: _____

5. Select *all* the scales that are equivalent to 4 centimeters to 20 meters.

- ☐ A. 4 inches to 20 inches
☒ B. 1 centimeter to 5 meters
☒ C. 10 meters to 2 centimeters
☐ D. 5 millimeters to 3 meters
☐ E. 1 inch to 4 feet

Explain your thinking for the scale(s) you selected.

Explanations vary. Since the scale is 4 centimeters to 20 meters, I can divide both sides by 4 to get an equivalent scale of 1 centimeter to 5 meters. I can also divide both sides of the scale by 2 and flip the order, to get the equivalent scale 10 meters to 2 centimeters.

6. On a blueprint, the living room is 4.2 inches wide. The blueprint has a scale of 1 inch to 5 feet. How wide would the living room be on a blueprint that has a scale of 1 inch to 20 feet?

- A. 1.2 inches
☒ B. 2.1 inches
C. 3.2 inches
D. 31 inches

Unit 1 Lesson 10

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	7.G.A.1
2	2	7.G.A.1
3	2	7.G.A.1
4	2	7.G.A.1
5	2	7.G.A.1
6	1	7.G.A.1

Notes:

Additional Practice

ACC7.3.10

1. Which is the solution to the equation $3x + 7 = -11$?

A. 9

B. 6

C. -6

D. -9

2. Which is the solution to the equation $\frac{2}{3}(x + 8) = -4$?

A. 14

B. 2

C. -2

D. -14

Problems 3–6: Solve each equation. Show your thinking.

3. $41 = 11 + 4z$

$z = 7\frac{1}{2}$. *Work varies.*

$41 = 11 + 4z$

$30 = 4z$

$7\frac{1}{2} = z$

4. $3w + 3.1 = 9.7$

$w = 2.2$. *Work varies.*

$3w + 3.1 = 9.7$

$3w = 6.6$

$w = 2.2$

5. $-\frac{5}{4}(x - 6) = \frac{10}{4}$

$x = 4$. *Work varies.*

$-\frac{5}{4}(x - 6) = \frac{10}{4}$

$x - 6 = -2$

$x = 4$

6. $8000 = 4000(x - 0.06)$

$x = 2.06$. *Work varies.*

$8000 = 4000(x - 0.06)$

$2 = x - 0.06$

$2.06 = x$

7. Two students are solving the equation $4(x + 6) = 22$. One student starts by using the distributive property. Another student starts by dividing both sides by 4. Show what each student's solution might look like.

Distributive Property First

$$4(x + 6) = 22$$

$$4x + 24 = 22$$

$$4x = -2$$

$$x = -\frac{1}{2} \text{ (or equivalent)}$$

Dividing First

$$4(x + 6) = 22$$

$$x + 6 = \frac{22}{4}$$

$$x = \frac{22}{4} - 6$$

$$x = -\frac{1}{2} \text{ (or equivalent)}$$

8. Solve the equation $-\frac{3}{4}(x - 8) = -6$ using two different methods. Show your thinking.

Responses vary.

Method 1:

$$-\frac{3}{4}(x - 8) \div \left(-\frac{3}{4}\right) = -6 \div \left(-\frac{3}{4}\right)$$

$$x - 8 = 8$$

$$x - 8 + 8 = 8 + 8$$

$$x = 16$$

Method 2:

$$-\frac{3}{4}x + 6 = -6$$

$$-\frac{3}{4}x + 6 - 6 = -6 - 6$$

$$-\frac{3}{4}x = -12$$

$$-\frac{3}{4}x \div \left(-\frac{3}{4}\right) = -12 \div \left(-\frac{3}{4}\right)$$

$$x = 16$$

9. Diego solved the equation $-3(x + 6) = 21$. His response is shown. Diego made a mistake. Identify his mistake, and then correct his response. Show or explain your thinking.

When solving the equation, Diego incorrectly distributed the -3 to the terms in parentheses. $-3 \cdot 6$ is -18 .

$$-3(x + 6) = 21$$

$$-3x - 18 = 21$$

$$-3x = 39$$

$$x = -13$$

$$-3(x + 6) = 21$$

$$-3x + 18 = 21$$

$$-3x = 3$$

$$x = -1$$

Name: _____ Date: _____ Period: _____

Additional Practice

ACC7.3.10

1. Which is the solution to the equation $3x + 7 = -11$?

A. 9
B. 6
C. -6
D. -9

2. Which is the solution to the equation $\frac{2}{3}(x + 8) = -4$?

A. 14
B. 2
C. -2
D. -14

Problems 3–6: Solve each equation. Show your thinking.

3. $41 = 11 + 4z$

$z = 7\frac{1}{2}$. Work varies.
 $41 = 11 + 4z$
 $30 = 4z$
 $7\frac{1}{2} = z$

4. $3w + 3.1 = 9.7$

$w = 2.2$. Work varies.
 $3w + 3.1 = 9.7$
 $3w = 6.6$
 $w = 2.2$

5. $-\frac{5}{4}(x - 6) = \frac{10}{4}$
 $x = 4$. Work varies.
 $-\frac{5}{4}(x - 6) = \frac{10}{4}$
 $x - 6 = -2$
 $x = 4$

6. $8000 = 4000(x - 0.06)$
 $x = 2.06$. Work varies.
 $8000 = 4000(x - 0.06)$
 $2 = x - 0.06$
 $2.06 = x$

Unit 3 Lesson 10

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Name: _____ Date: _____ Period: _____

7. Two students are solving the equation $4(x + 6) = 22$. One student starts by using the distributive property. Another student starts by dividing both sides by 4. Show what each student's solution might look like.

Distributive Property First

$4(x + 6) = 22$
 $4x + 24 = 22$
 $4x = -2$
 $x = -\frac{1}{2}$ (or equivalent)

Dividing First

$4(x + 6) = 22$
 $x + 6 = \frac{22}{4}$
 $x + 6 = \frac{11}{2}$
 $x = \frac{11}{2} - 6$
 $x = -\frac{1}{2}$ (or equivalent)

8. Solve the equation $-\frac{3}{4}(x - 8) = -6$ using two different methods. Show your thinking.

Responses vary.

Method 1:

$-\frac{3}{4}(x - 8) \div (-\frac{3}{4}) = -6 \div (-\frac{3}{4})$
 $x - 8 = 8$
 $x - 8 + 8 = 8 + 8$
 $x = 16$

Method 2:

$-\frac{3}{4}x + 6 = -6$
 $-\frac{3}{4}x + 6 - 6 = -6 - 6$
 $-\frac{3}{4}x = -12$
 $-\frac{3}{4}x \div (-\frac{3}{4}) = -12 \div (-\frac{3}{4})$
 $x = 16$

9. Diego solved the equation $-3(x + 6) = 21$. His response is shown. Diego made a mistake. Identify his mistake, and then correct his response. Show or explain your thinking.

When solving the equation, Diego incorrectly distributed the -3 to the terms in parentheses. $-3 \cdot 6$ is -18 .

$-3(x + 6) = 21$
 $-3x - 18 = 21$
 $-3x = 39$
 $x = -13$

$-3(x + 6) = 21$

$-3x + 18 = 21$

$-3x = 3$

$x = -1$

Unit 3 Lesson 10

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	7.EE.B.4.A
2	1	7.EE.B.4.A
3	2	7.EE.B.4.A
4	2	7.EE.B.4.A
5	2	7.EE.B.4.A
6	2	7.EE.B.4.A
7	2	7.EE.B.4.A
8	2	7.EE.B.4.A
9	2	7.EE.B.4.A

Notes:

Additional Practice

ACC7.3.11

1. Elena had 100 fliers to pass out. She gave an equal amount of fliers to 5 volunteers and she passed out 10 fliers. Which equation represents this situation?

A. $100 = 5(x + 10)$

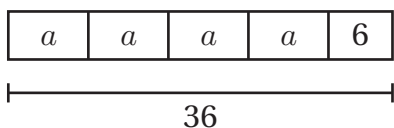
B. $10(x + 5) = 100$

C. $10x + 5 = 100$

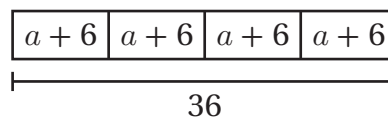
D. $100 = 5x + 10$

2. Which tape diagram represents the equation $4(a + 6) = 36$? **Tape Diagram B**

Tape Diagram A



Tape Diagram B



3. Match each scenario with the equation that it could represent.

Equation

Scenario

a. $4(x + 2) = 18$

b Jada has 4 dimes and 2 times as many quarters as Han. She has 18 dimes and quarters altogether. How many quarters does Han have?

b. $2x + 4 = 18$

a Lin's father spent \$18 and purchased each of his 4 children a \$2 yo-yo and a bottle of bubbles. How much does each bottle of bubbles cost?

c. $4x + 2 = 18$

d Diego made two fruit baskets using 18 pieces of fruit. He placed an equal number of pieces in each basket and then added 4 more pieces to each basket. How many pieces of fruit did Diego originally put in each basket?

d. $2(x + 4) = 18$

c Clare has 18 gel pens. She gives an equal amount of gel pens to 4 friends and keeps 2 for herself. How many gel pens did she give each friend?

Problems 4–6: For each scenario, write and solve an equation. Be sure to identify what the variable represents. Then explain the solution's meaning in context. Draw a tape diagram if it helps with your thinking.

4. Andre tutored 8 times this past month and earned the same amount each time he tutored. To thank him, the family gave him an extra \$6 at the end of the month. Andre earned \$110 from tutoring.

a Equation, variable meaning, and solution:
 $8e + 6 = 110$, $e = 13$, e represents the amount Andre earned each time he tutored.

b Solution's Meaning:
Andre earned \$13 each time he tutored.

5. Mai spent 5 more minutes reading than Clare. Lin read 3 times as many minutes as Mai. If Lin read for 30 minutes, how many minutes did Clare spend reading?

a Equation, variable meaning, and solution:
 $3(x + 5) = 30$, $x = 5$, x represents the number of minutes Clare spent reading.

b Solution's Meaning:
Clare read 5 minutes.

6. Priya and Tyler both run every day. Priya runs 1.5 miles less than Tyler each day. If after 4 days Priya has run a total of 20 miles, how far does Tyler run each day?

a Equation, variable meaning, and solution:
 $4(m - 1.5) = 20$, $m = 6.5$, m represents the number of miles Tyler runs each day.

b Solution's Meaning:
Tyler runs 6.5 miles each day.

7. Write a scenario that could be represented by each of the following equations.

a $3(x - 7) = 21$
Responses vary. Bard scored 7 points less than Shawn in the basketball game. Andre scored three times as many points as Bard. If Andre scored 21 points, how many points did Shawn score?

b $2y - 20 = -5$
Responses vary. Priya has an allowance balance of $-\$20$, due to overspending. In two weeks, she makes 2 equal payments to her mother and now has a balance of $-\$5$. How much did she pay her mom each time?

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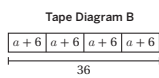
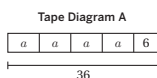
Additional Practice

ACC7.3.11

1. Elena had 100 fliers to pass out. She gave an equal amount of fliers to 5 volunteers and she passed out 10 fliers. Which equation represents this situation?

- A. $100 = 5(x + 10)$ B. $10(x + 5) = 100$
C. $10x + 5 = 100$ **D. $100 = 5x + 10$**

2. Which tape diagram represents the equation $4(a + 6) = 36$? **Tape Diagram B**



3. Match each scenario with the equation that it could represent.

- | Equation | Scenario |
|--------------------|---|
| a. $4(x + 2) = 18$ | b Jada has 4 dimes and 2 times as many quarters as Han. She has 18 dimes and quarters altogether. How many quarters does Han have? |
| b. $2x + 4 = 18$ | a Lin's father spent \$18 and purchased each of his 4 children a \$2 yo-yo and a bottle of bubbles. How much does each bottle of bubbles cost? |
| c. $4x + 2 = 18$ | d Diego made two fruit baskets using 18 pieces of fruit. He placed an equal number of pieces in each basket and then added 4 more pieces to each basket. How many pieces of fruit did Diego originally put in each basket? |
| d. $2(x + 4) = 18$ | c Clare has 18 gel pens. She gives an equal amount of gel pens to 4 friends and keeps 2 for herself. How many gel pens did she give each friend? |

Unit 3 Lesson 11

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Problems 4–6: For each scenario, write and solve an equation. Be sure to identify what the variable represents. Then explain the solution's meaning in context. Draw a tape diagram if it helps with your thinking.

4. Andre tutored 8 times this past month and earned the same amount each time he tutored. To thank him, the family gave him an extra \$6 at the end of the month. Andre earned \$110 from tutoring.

- a** Equation, variable meaning, and solution:
 $8x + 6 = 110$, $x = 13$, x represents the amount Andre earned each time he tutored.
b Solution's Meaning:
Andre earned \$13 each time he tutored.

5. Mai spent 5 more minutes reading than Clare. Lin read 3 times as many minutes as Mai. If Lin read for 30 minutes, how many minutes did Clare spend reading?

- a** Equation, variable meaning, and solution:
 $3(x + 5) = 30$, $x = 5$, x represents the number of minutes Clare spent reading.
b Solution's Meaning:
Clare read 5 minutes.

6. Priya and Tyler both run every day. Priya runs 1.5 miles less than Tyler each day. If after 4 days Priya has run a total of 20 miles, how far does Tyler run each day?

- a** Equation, variable meaning, and solution:
 $4(m - 1.5) = 20$, $m = 6.5$, m represents the number of miles Tyler runs each day.
b Solution's Meaning:
Tyler runs 6.5 miles each day.

7. Write a scenario that could be represented by each of the following equations.

- a** $3(x - 7) = 21$
Responses vary. Bard scored 7 points less than Shawn in the basketball game. Andre scored three times as many points as Bard. If Andre scored 21 points, how many points did Shawn score?
b $2y - 20 = -5$
Responses vary. Priya has an allowance balance of -\$20, due to overspending. In two weeks, she makes 2 equal payments to her mother and now has a balance of -\$5. How much did she pay her mom each time?

Unit 3 Lesson 11

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Additional Practice

Practice Problem Analysis

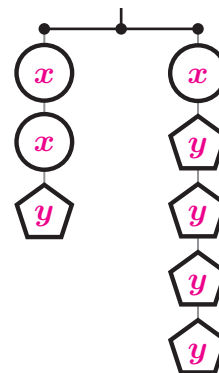
Problem	DOK	Standard(s)
1	2	7.EE.B.4
2	2	7.EE.B.4
3	2	7.EE.B.4
4	2	7.EE.B.4.A
5	2	7.EE.B.4.A
6	2	7.EE.B.4.A
7	3	7.EE.B.4.A

Notes:

Additional Practice

ACC7.3.12

Problems 1–2: In this balanced hanger diagram, the weight of the circle is x and the weight of the pentagon is y .



1. Label each figure with x or y . Then, write an equation using x and y to represent the hanger diagram.

$$x + x + y = x + y + y + y + y \text{ (or equivalent)}$$

2. If $x = 9$, what is the value of y ? Show or explain your thinking.

$y = 3$. *Work varies.*

$$9 + 9 + y = 9 + y + y + y + y$$

$$18 + y = 9 + 4y$$

$$9 = 3y$$

$$3 = y$$

3. In this balanced hanger diagram, the weight of the triangle is x and the weight of the circle is y .

a Label each figure with x or y .

b Select *all* the equations that could represent a balanced hanger.

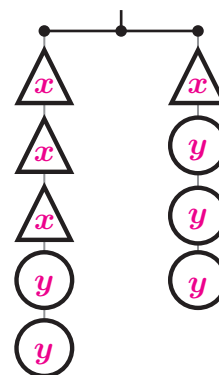
☐ A. $3y + 2x = y + 3x$

☒ B. $3x + 2y = x + 3y$

☒ C. $2x + 2y = 3y$

☐ D. $5xy = 4xy$

☒ E. $2x = y$



4. Match each set of equations with a possible step that turns the first equation into the second equation.

Equation	Possible steps
a. $9x + 3 = 3x + 18$ $3x + 1 = x + 6$	 c Multiply each side by -3 .
b. $-3(5x - 7) = -21$ $5x - 7 = 7$	 b Divide each side by -3 .
c. $-\frac{4}{3}x = 5$ $4x = -15$	 a Divide each side by 3 .
d. $7x + 5 = 3x - 6$ $4x + 5 = -6$	 e Subtract 3 from each side.
e. $10 - 5x = 8 + 4x$ $7 - 5x = 5 + 4x$	 d Subtract $3x$ from each side.

Problems 5–6: Jada and Tyler were each trying to solve $7x - 4 = 5x - 5$. Describe the first step they each made to the equation.

5. The result of Jada's first step was $7x = 5x - 1$.

Jada added 4 to each side.

6. The result of Tyler's first step was $7x + 1 = 5x$.

Tyler added 5 to each side.

Name: _____ Date: _____ Period: _____

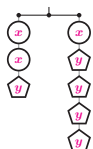
Additional Practice

ACC7.3.12

Problems 1–2: In this balanced hanger diagram, the weight of the circle is x and the weight of the pentagon is y .

1. Label each figure with x or y . Then, write an equation using x and y to represent the hanger diagram.

$x + x + y = x + y + y + y$ (or equivalent)



2. If $x = 9$, what is the value of y ? Show or explain your thinking.

$y = 3$. Work varies.

$9 + 9 + y = 9 + y + y + y$

$18 + y = 9 + 4y$

$9 = 3y$

$3 = y$

3. In this balanced hanger diagram, the weight of the triangle is x and the weight of the circle is y .

- a. Label each figure with x or y .

- b. Select all the equations that could represent a balanced hanger.

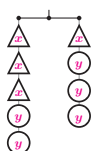
☐ A. $3y + 2x = y + 3x$

☒ B. $3x + 2y = x + 3y$

☒ C. $2x + 2y = 3y$

☐ D. $5xy = 4xy$

☒ E. $2x = y$



Unit 3 Lesson 12

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4. Match each set of equations with a possible step that turns the first equation into the second equation.

Equation

Possible steps

a. $9x + 3 = 3x + 18$
 $3x + 1 = x + 6$

c. Multiply each side by -3 .

b. $-3(5x - 7) = -21$
 $5x - 7 = 7$

b. Divide each side by -3 .

c. $-\frac{4}{3}x = 5$
 $4x = -15$

a. Divide each side by 3.

d. $7x + 5 = 3x - 6$
 $4x + 5 = -6$

e. Subtract 3 from each side.

e. $10 - 5x = 8 + 4x$
 $7 - 5x = 5 + 4x$

d. Subtract $3x$ from each side.

Problems 5–6: Jada and Tyler were each trying to solve $7x - 4 = 5x - 5$. Describe the first step they each made to the equation.

5. The result of Jada's first step was $7x = 5x - 1$.

Jada added 4 to each side.

6. The result of Tyler's first step was $7x + 1 = 5x$.

Tyler added 5 to each side.

Unit 3 Lesson 12

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.EE.C.7
2	2	8.EE.C.7
3	2	8.EE.C.7
4	2	8.EE.C.7
5	2	8.EE.C.7
6	2	8.EE.C.7

Notes:

Additional Practice

ACC7.3.13

1. Andre and Clare are each solving the equation $-4(x - 5) = 10x - 8$. Andre's solution is $x = -2$, and Clare's solution is $x = -\frac{7}{9}$. Their responses are shown.

Andre's solution

$$\begin{aligned} -4(x - 5) &= 10x - 8 \\ 4x - 20 &= 10x - 8 \\ -20 &= 6x - 8 \\ -12 &= 6x \\ -2 &= x \end{aligned}$$

Clare's solution

$$\begin{aligned} -4(x - 5) &= 10x - 8 \\ x - 5 &= 10x + 2 \\ -5 &= 9x + 2 \\ -7 &= 9x \\ -\frac{7}{9} &= x \end{aligned}$$

Whose solution is correct?

Andre

Clare

Neither

Show or explain your thinking.

Explanations vary. Andre's solution is incorrect because he distributed incorrectly on the second line. Clare's response is also incorrect because when dividing both sides of the equation by -4 , she didn't divide $10x$ by -4 .

2. Solve the equation $-5(2x - 2) = 20x$ two different ways. Show your thinking.

Responses vary.

Method 1:

$$\begin{aligned} -5(2x - 2) &= 20x \\ 2x - 2 &= -4x \\ 6x &= 2 \\ x &= \frac{1}{3} \end{aligned}$$

Method 2:

$$\begin{aligned} -5(2x - 2) &= 20x \\ -10x + 10 &= 20x \\ 10 &= 30x \\ \frac{1}{3} &= x \end{aligned}$$

3. Mai solved the equation $\frac{1}{2}(6x - 8) = 5x + 12$. Mai began with the original equation and her steps are shown. Complete the table by describing the steps Mai used to solve the equation.

Steps	Description
$\frac{1}{2}(6x - 8) = 5x + 12$	Original equation
$3x - 4 = 5x + 12$	Distribute $\frac{1}{2}$ into the expression on the left.
$-4 = 2x + 12$	Subtract $3x$ from each side.
$-16 = 2x$	Subtract 12 from each side.
$-8 = x$	Divide both sides by 2.

Problems 4–5: Determine whether $x = 2$ is a solution for each equation. Show your thinking.

4. $4(x - 5) = -12$
Yes. Work varies.

$$\begin{aligned} 4(2 - 5) &= -12 \\ 4(-3) &= -12 \\ -12 &= -12 \end{aligned}$$

5. $-3(x + 2) = 2x + 16$
No. Work varies.

$$\begin{aligned} -3(2 + 2) &= 2(2) + 16 \\ -3(4) &= 4 + 16 \\ -12 &= 20 \end{aligned}$$

6. Solve the equation. Show your thinking. Check your solution.

$$\begin{aligned} 4(x + 3) &= 6x + 2(x - 4) \\ x &= 5. \text{ Work varies.} \end{aligned}$$

$$\begin{aligned} 4x + 12 &= 6x + 2x - 8 \\ 4x + 12 &= 8x - 8 \\ 20 &= 4x \\ 5 &= x \end{aligned}$$

Check:

$$\begin{aligned} 4(5 + 3) &= 6(5) + 2(5 - 4) \\ 4(8) &= 30 + 2(1) \\ 32 &= 30 + 2 \\ 32 &= 32 \end{aligned}$$

Name: _____ Date: _____ Period: _____

Additional Practice

ACC7.3.13

1. Andre and Clare are each solving the equation $-4(x - 5) = 10x - 8$. Andre's solution is $x = -2$, and Clare's solution is $x = -\frac{7}{9}$. Their responses are shown.

Andre's solution	Clare's solution
$-4(x - 5) = 10x - 8$	$-4(x - 5) = 10x - 8$
$4x - 20 = 10x - 8$	$x - 5 = 10x + 2$
$-20 = 6x - 8$	$-5 = 9x + 2$
$-12 = 6x$	$-7 = 9x$
$-2 = x$	$-\frac{7}{9} = x$

Whose solution is correct?

Andre Clare **Neither**

Show or explain your thinking.

Explanations vary. Andre's solution is incorrect because he distributed incorrectly on the second line. Clare's response is also incorrect because when dividing both sides of the equation by -4 , she didn't divide $10x$ by -4 .

2. Solve the equation $-5(2x - 2) = 20x$ two different ways. Show your thinking.

Responses vary.

Method 1:
 $-5(2x - 2) = 20x$
 $2x - 2 = -4x$
 $6x = 2$
 $x = \frac{1}{3}$

Method 2:
 $-5(2x - 2) = 20x$
 $-10x + 10 = 20x$
 $10 = 30x$
 $\frac{1}{3} = x$

Unit 3 Lesson 13

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Name: _____ Date: _____ Period: _____

3. Mai solved the equation $\frac{1}{2}(6x - 8) = 5x + 12$. Mai began with the original equation and her steps are shown. Complete the table by describing the steps Mai used to solve the equation.

Steps	Description
$\frac{1}{2}(6x - 8) = 5x + 12$	Original equation
$3x - 4 = 5x + 12$	Distribute $\frac{1}{2}$ into the expression on the left.
$-4 = 2x + 12$	Subtract $3x$ from each side.
$-16 = 2x$	Subtract 12 from each side.
$-8 = x$	Divide both sides by 2.

Problems 4–5: Determine whether $x = 2$ is a solution for each equation. Show your thinking.

4. $4(x - 5) = -12$

Yes. Work varies.

$$4(2 - 5) = -12$$

$$4(-3) = -12$$

$$-12 = -12$$

5. $-3(x + 2) = 2x + 16$

No. Work varies.

$$-3(2 + 2) = 2(2) + 16$$

$$-3(4) = 4 + 16$$

$$-12 = 20$$

6. Solve the equation. Show your thinking. Check your solution.

$$4(x + 3) = 6x + 2(x - 4)$$

$$x = 5. \text{ Work varies.}$$

$$4x + 12 = 6x + 2x - 8$$

$$4x + 12 = 8x - 8$$

$$20 = 4x$$

$$5 = x$$

Check:

$$4(5 + 3) = 6(5) + 2(5 - 4)$$

$$4(8) = 30 + 2(1)$$

$$32 = 30 + 2$$

$$32 = 32$$

Unit 3 Lesson 13

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	8.EE.C.7.B
2	2	8.EE.C.7.B
3	2	8.EE.C.7.B
4	1	8.EE.C.7.B
5	1	8.EE.C.7.B
6	1	8.EE.C.7.B

Notes:

Additional Practice

ACC7.3.15

1. For each equation, determine whether it has *one solution*, *no solution*, or *infinitely many solutions* by placing a check mark in the appropriate box.

Equation	One Solution	No Solution	Infinitely Many Solutions
$x + 2 = x + 4$		✓	
$2x + 4 = 2x + 1 + 3$			✓
$3(x - 1) = 2x + 1$	✓		
$-(4x - 5) = -x + 5 - 3x$			✓

2. If an equation is never true for any value of x , which is true about the equation?

- A.** It has no solution.
B. It has one solution.
C. It has infinitely many solutions.
D. Zero is its only solution.

Problems 3–4: For each equation, decide whether it has *one solution*, *no solution*, or *infinitely many solutions*. Show or explain your thinking.

3. $3x - 7x + 1 = -4x + 5$

No solution. Explanations vary.
 By combining like terms on the left side, the new equation is $-4x + 1 = -4x + 5$. The expressions on either side will never be true for any value of x .

4. $4(x + 3) = -2(2x + 6)$

One solution. Explanations vary.
 $4x + 12 = -4x - 12$
 $8x + 12 = -12$
 $8x = -24$
 $x = -3$

5. Elena said $4x - 6 = 4(x - 6)$ has infinitely many solutions. Is Elena correct? Explain your thinking.

No. Explanations vary. Distributing on the right side and subtracting $4x$ from both sides leaves the false statement: $-6 = -24$. This means that there is no value of x that will make this a true statement.

6. Noah and Lin solved the equation $-(3x + 2) - 1 = -2x - (x + 3)$. Their responses are shown below.

Noah's Work

$$\begin{aligned} -(3x + 2) - 1 &= -2x - (x + 3) \\ -3x - 2 - 1 &= -2x - x - 3 \\ -3x - 3 &= -3x - 3 \\ -3 &= -3 \end{aligned}$$

Lin's Work

$$\begin{aligned} -(3x + 2) - 1 &= -2x - (x + 3) \\ -3x - 2 - 1 &= -2x - x - 3 \\ -3x - 3 &= -3x - 3 \\ -3x &= -3x \\ x &= x \end{aligned}$$

Noah says that one of them made a mistake because you cannot get different results for the same equation. Is Noah correct? Explain your thinking.

Noah's statement is not correct. Explanations vary. Both Noah and Lin ended up with a true statement. There are infinitely many solutions for this equation.

7. Write the other side of this equation so that it is true for all values of x .

$$-\frac{1}{2}(4x + 12) + 5x = \dots$$

$3x - 6$ (or equivalent)

8. Write the other side of this equation so that it is true for no values of x .

$$-\frac{1}{2}(4x + 12) + 5x = \dots$$

Responses vary. $3x + 2$

9. Write the other side of this equation so that it is true when $x = 0$.

$$-\frac{1}{2}(4x + 12) + 5x = \dots$$

Responses vary. $4x - 6$

Name: _____ Date: _____ Period: _____

Additional Practice

ACC7.3.15

1. For each equation, determine whether it has one solution, no solution, or infinitely many solutions by placing a check mark in the appropriate box.

Equation	One Solution	No Solution	Infinitely Many Solutions
$x + 2 = x + 4$		✓	
$2x + 4 = 2x + 1 + 3$			✓
$3(x - 1) = 2x + 1$	✓		
$-(4x - 5) = -x + 5 - 3x$			✓

2. If an equation is never true for any value of x , which is true about the equation?

- ☒ A. It has no solution.
☐ B. It has one solution.
☐ C. It has infinitely many solutions.
☐ D. Zero is its only solution.

Problems 3–4: For each equation, decide whether it has one solution, no solution, or infinitely many solutions. Show or explain your thinking.

3. $3x - 7x + 1 = -4x + 5$

No solution. *Explanations vary.* By combining like terms on the left side, the new equation is $-4x + 1 = -4x + 5$. The expressions on either side will never be true for any value of x .

4. $4(x + 3) = -2(2x + 6)$

One solution. *Explanations vary.*
 $4x + 12 = -4x - 12$
 $8x + 12 = -12$
 $8x = -24$
 $x = -3$

5. Elena said $4x - 6 = 4(x - 6)$ has infinitely many solutions. Is Elena correct? Explain your thinking.

No. *Explanations vary.* Distributing on the right side and subtracting $4x$ from both sides leaves the false statement: $-6 = -24$. This means that there is no value of x that will make this a true statement.

Unit 3 Lesson 15

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6. Noah and Lin solved the equation $-(3x + 2) - 1 = -2x - (x + 3)$. Their responses are shown below.

Noah's Work

$$\begin{aligned} -(3x + 2) - 1 &= -2x - (x + 3) \\ -3x - 2 - 1 &= -2x - x - 3 \\ -3x - 3 &= -3x - 3 \\ -3 &= -3 \end{aligned}$$

Lin's Work

$$\begin{aligned} -(3x + 2) - 1 &= -2x - (x + 3) \\ -3x - 2 - 1 &= -2x - x - 3 \\ -3x - 3 &= -3x - 3 \\ -3x &= -3x \\ x &= x \end{aligned}$$

Noah says that one of them made a mistake because you cannot get different results for the same equation. Is Noah correct? Explain your thinking.

Noah's statement is not correct. *Explanations vary.* Both Noah and Lin ended up with a true statement. There are infinitely many solutions for this equation.

7. Write the other side of this equation so that it is true for all values of x .

$$-\frac{1}{2}(4x + 12) + 5x = \underline{\hspace{2cm}}$$

Responses vary. $3x - 6$ (or equivalent)

8. Write the other side of this equation so that it is true for no values of x .

$$-\frac{1}{2}(4x + 12) + 5x = \underline{\hspace{2cm}}$$

Responses vary. $3x + 2$

9. Write the other side of this equation so that it is true when $x = 0$.

$$-\frac{1}{2}(4x + 12) + 5x = \underline{\hspace{2cm}}$$

Responses vary. $4x - 6$

Unit 3 Lesson 15

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.EE.C.7.A
2	1	8.EE.C.7.A
3	2	8.EE.C.7.A
4	2	8.EE.C.7.A
5	1	8.EE.C.7.A
6	2	8.EE.C.7.A
7	2	8.EE.C.7.A
8	2	8.EE.C.7.A
9	2	8.EE.C.7.A

Notes:

Additional Practice

ACC7.3.16

1. Without solving them, identify whether each equation has a solution that is *positive*, *negative*, or *zero* by placing a check mark in the appropriate box.

Equation	Positive	Negative	Zero
$12x = -36.24$		✓	
$2x = 14.6$	✓		
$6x - 10 = -10$			✓
$-\frac{2}{3}x = -64$	✓		

Problems 2–5: Solve each equation. Show your thinking.

2. $5x + 7x - 12 = 6(2x - 2)$

Infinitely many solutions

3. $\frac{1}{2}(6y - 12) = 2(y - \frac{3}{2})$

$3y - 6 = 2y - 3$

$y - 6 = -3$

$y = 3$

4. $4(6 - 2m) = 3(m + 1) + 10$

$24 - 8m = 3m + 3 + 10$

$24 - 11m = 13$

$-11m = -11$

$m = 1$

5. $6b + 8 - 10b + 4 = -16 - 4b - 2$

No solution

6. Andre studied the equation $3(3x - 6) = -9(x - 2)$. He said, "I can tell right away there is no solution because on the left side, you will have $9x$, and on the right side, you will have $-9x$. $-9x$ and $9x$ make zero."

Is Andre correct? Explain your thinking.

No. Explanations vary. The $9x$ and $-9x$ will add up to either $-18x$ or $18x$, depending on which side of the equation you collect them. There will be one solution to this equation.

Problems 7–8: Jada wrote the equation $8x + 2 = 8x + 4$. She wants to change only one term so that the equation has exactly one solution.

7. What is an example of a change that would result in exactly one solution?

Responses vary. If you change one of the $8x$'s to any term with x except for positive $8x$, this will result in an equation with one solution.

8. What is an example of a change that would result in an *infinite number* of solutions?

Responses vary. If you change the constant terms of 2 and 4 to the same number, this will result in an equation with an infinite number of solutions.

Name: _____ Date: _____ Period: _____

Additional Practice

ACC7.3.16

1. Without solving them, identify whether each equation has a solution that is positive, negative, or zero by placing a check mark in the appropriate box.

Equation	Positive	Negative	Zero
$12x = -36.24$		✓	
$2x = 14.6$	✓		
$6x - 10 = -10$			✓
$-\frac{2}{3}x = -64$	✓		

Problems 2–5: Solve each equation. Show your thinking.

2. $5x + 7x - 12 = 6(2x - 2)$
Infinitely many solutions

3. $\frac{3}{4}(6y - 12) = 2(y - \frac{3}{2})$
 $3y - 6 = 2y - 3$
 $y - 6 = -3$
 $y = 3$

4. $4(6 - 2m) = 3(m + 1) + 10$
 $24 - 8m = 3m + 3 + 10$
 $24 - 11m = 13$
 $-11m = -11$
 $m = 1$

5. $6b + 8 - 10b + 4 = -16 - 4b - 2$
No solution

Unit 3 Lesson 16

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Name: _____ Date: _____ Period: _____

6. Andre studied the equation $3(3x - 6) = -9(x - 2)$. He said, "I can tell right away there is no solution because on the left side, you will have $9x$, and on the right side, you will have $-9x$. $-9x$ and $9x$ make zero."

Is Andre correct? Explain your thinking.

No. Explanations vary. The $9x$ and $-9x$ will add up to either $-18x$ or $18x$, depending on which side of the equation you collect them. There will be one solution to this equation.

Problems 7–8: Jada wrote the equation $8x + 2 = 8x + 4$. She wants to change only one term so that the equation has exactly one solution.

7. What is an example of a change that would result in exactly one solution?
Responses vary. If you change one of the $8x$'s to any term with x except for positive $8x$, this will result in an equation with one solution.

8. What is an example of a change that would result in an infinite number of solutions?
Responses vary. If you change the constant terms of 2 and 4 to the same number, this will result in an equation with an infinite number of solutions.

Unit 3 Lesson 16

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.EE.C.7.A
2	1	8.EE.C.7.A
3	1	8.EE.C.7.A
4	1	8.EE.C.7.A
5	1	8.EE.C.7.A
6	2	8.EE.C.7.A
7	2	8.EE.C.7.A
8	2	8.EE.C.7.A

Notes:

Additional Practice

ACC7.3.17

1. For what value of x do the expressions $\frac{3}{4}x + 12$ and $\frac{1}{4}x - 8$ have the same value? Show or explain your thinking.

–40. *Work varies.*

$$\begin{aligned}\frac{3}{4}x + 12 &= \frac{1}{4}x - 8 \\ 4\left(\frac{3}{4}x + 12\right) &= 4\left(\frac{1}{4}x - 8\right) \\ 3x + 48 &= x - 32 \\ 2x + 48 &= -32 \\ 2x &= -80 \\ x &= -40\end{aligned}$$

2. Which story could the equation $4x + 4 = 5x - 6$ represent?

- ☒ A. Han and his sibling get a weekly allowance of x dollars. At one point, Han had an allowance balance of $-\$6.00$ and his sibling had an allowance balance of $\$4.00$. After 5 weeks of allowance for Han and 4 weeks of allowance for his sibling, they have the same allowance balance.
- B. The Huskies and the Cardinals are playing a trivia game. Each correct question is worth x points. At one point in the game, the Huskies have -6 points and the Cardinals have 4 points. After the Huskies answer 4 correct questions and the Cardinals answer 5 correct questions, they have the same number of points.

3. At Lin's fitness center, a membership costs $\$40$ a month and there is a one-time registration fee of $\$20$. At Kiran's fitness center, a membership costs $\$50$ a month and there is a one-time registration fee of $\$10$.

Which equation represents when the costs of the membership would be equal if m represents the number of months?

- A. $40m + 10 = 50m + 20$
- B. $40m = 50m + 10$
- C. $40m + 20 = 50m$
- ☒ D. $40m + 20 = 50m + 10$

Problems 4–7: Clare and Mai are running in the same direction on the same running trail.

4. Clare runs at a constant speed of 4 miles per hour. Write an expression that shows how many miles Clare has run after t hours.

$$4t$$

5. Mai started running $\frac{1}{4}$ of an hour after Clare. If Clare has been running for t hours, how long has Mai been running?

$$t - \frac{1}{4} \text{ (or equivalent)}$$

6. Mai runs at a constant speed of 6 miles per hour. Write an expression that shows how many miles Mai has run after Clare has been running for t hours.

$$6\left(t - \frac{1}{4}\right) \text{ miles (or equivalent)}$$

7. Use your expressions to determine when Clare and Mai will meet each other on the running trail. Show or explain your thinking.

$\frac{3}{4}$ hours. *Explanations vary.*

$$4t = 6\left(t - \frac{1}{4}\right)$$

$$4t = 6t - \frac{3}{2}$$

$$-2t = -\frac{3}{2}$$

$$-2(-2t) = -2\left(-\frac{3}{2}\right)$$

$$4t = 3$$

$$t = \frac{3}{4}$$

Claire and Mai will meet on the trail after $\frac{3}{4}$ of an hour.

8. Shawn wants to take photography lessons. The instructor offers two options. Option 1 costs \$60 per lesson and comes with a camera. Option 2 costs \$40 per lesson but you have to purchase your own camera. Suppose Shawn buys a \$200 camera and chooses Option 2. After how many lessons, x , is Shawn's total cost the same as it would have been if he had chosen Option 1? Show your thinking.

10 lessons. *Explanations vary.*

$$60x = 200 + 40x$$

$$20x = 200$$

$$x = 10$$

After 10 lessons, the cost of Option 1 and Option 2 will be the same.

Name: _____ Date: _____ Period: _____

Additional Practice

ACC7.3.17

1. For what value of x do the expressions $\frac{3}{4}x + 12$ and $\frac{1}{4}x - 8$ have the same value? Show or explain your thinking.

–40. Work varies.

$$\begin{aligned}\frac{3}{4}x + 12 &= \frac{1}{4}x - 8 \\ 4\left(\frac{3}{4}x + 12\right) &= 4\left(\frac{1}{4}x - 8\right) \\ 3x + 48 &= x - 32 \\ 2x &= -80 \\ x &= -40\end{aligned}$$

2. Which story could the equation $4x + 4 = 5x - 6$ represent?

- (A) Han and his sibling get a weekly allowance of x dollars. At one point, Han had an allowance balance of $-\$6.00$ and his sibling had an allowance balance of $\$4.00$. After 5 weeks of allowance for Han and 4 weeks of allowance for his sibling, they have the same allowance balance.
- B. The Huskies and the Cardinals are playing a trivia game. Each correct question is worth x points. At one point in the game, the Huskies have -6 points and the Cardinals have 4 points. After the Huskies answer 4 correct questions and the Cardinals answer 5 correct questions, they have the same number of points.

3. At Lin's fitness center, a membership costs $\$40$ a month and there is a one-time registration fee of $\$20$. At Kiran's fitness center, a membership costs $\$50$ a month and there is a one-time registration fee of $\$10$.

Which equation represents when the costs of the membership would be equal if m represents the number of months?

- A. $40m + 10 = 50m + 20$
- B. $40m = 50m + 10$
- C. $40m + 20 = 50m$
- (D) $40m + 20 = 50m + 10$

Unit 3 Lesson 17

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Name: _____ Date: _____ Period: _____

Problems 4–7: Clare and Mai are running in the same direction on the same running trail.

4. Clare runs at a constant speed of 4 miles per hour. Write an expression that shows how many miles Clare has run after t hours.

$$4t$$

5. Mai started running $\frac{1}{4}$ of an hour after Clare. If Clare has been running for t hours, how long has Mai been running?

$$t - \frac{1}{4} \text{ (or equivalent)}$$

6. Mai runs at a constant speed of 6 miles per hour. Write an expression that shows how many miles Mai has run after Clare has been running for t hours.

$$6\left(t - \frac{1}{4}\right) \text{ miles (or equivalent)}$$

7. Use your expressions to determine when Clare and Mai will meet each other on the running trail. Show or explain your thinking.

$\frac{3}{4}$ hours. Explanations vary.

$$\begin{aligned}4t &= 6\left(t - \frac{1}{4}\right) \\ 4t &= 6t - \frac{3}{2} \\ -2t &= -\frac{3}{2} \\ -2(-2t) &= -2\left(-\frac{3}{2}\right) \\ 4t &= 3 \\ t &= \frac{3}{4}\end{aligned}$$

Claire and Mai will meet on the trail after $\frac{3}{4}$ of an hour.

8. Shawn wants to take photography lessons. The instructor offers two options. Option 1 costs $\$60$ per lesson and comes with a camera. Option 2 costs $\$40$ per lesson but you have to purchase your own camera. Suppose Shawn buys a $\$200$ camera and chooses Option 2. After how many lessons, x , is Shawn's total cost the same as it would have been if he had chosen Option 1? Show your thinking.

10 lessons. Explanations vary.

$$\begin{aligned}60x &= 200 + 40x \\ 20x &= 200 \\ x &= 10\end{aligned}$$

After 10 lessons, the cost of Option 1 and Option 2 will be the same.

Unit 3 Lesson 17

104

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	8.EE.C.7.B
2	1	8.EE.C.7.B
3	1	8.EE.C.7.B
4	1	8.EE.C.7.B
5	2	8.EE.C.7.B
6	1	8.EE.C.7.B
7	2	8.EE.C.7.B
8	2	8.EE.C.7.B

Notes:

Additional Practice

7.07

1. At a restaurant, a minimum of 2 people in each party need to be present in order to be seated at their table. Write an inequality that represents the possible number of people p that need to be present in order to be seated.

$p \geq 2$ (or equivalent)

2. Clare is younger than Bard. Bard is 13 years old. Write an inequality that compares Clare's age in years c to Bard's age.

$c < 13$ (or equivalent)

3. Diego started cooking dinner before 5:00 p.m. and finished cooking dinner at 6:00 p.m. Let d represent the number of hours Diego spent cooking dinner. Determine whether each statement is *definitely true*, *definitely not true*, or *possibly true*.

a $d < 1$

Definitely not true

c $d < 2$

Possibly true

e $d < 0.5$

Definitely not true

b $d > 1$

Definitely true

d $d > 2$

Possibly true

f $d > 0.5$

Definitely true

4. Noah started a run at 6:30 a.m. and finished sometime after 9:00 a.m. Let r represent the number of hours Noah spent running. Determine whether each statement is *definitely true*, *definitely not true*, or *possibly true*.

a $r > 2$

Definitely true

c $r > 3$

Possibly true

e $r < 2.5$

Definitely not true

b $r > 2.5$

Definitely true

d $r < 3$

Possibly true

f $r < 2$

Definitely not true

5. At the grocery store, all apples cost less than \$2.00 per pound.

- a What is the most expensive price one pound of apples could cost?

\$1.99

- b Write an inequality to represent the possible costs, in dollars, for any number of pounds of apples.

Sample response: $x < 2$, where x represents the cost, in dollars, for any number of pounds of apples

6. At the clothing store, all t -shirts cost more than \$9.99.

- a What is the least expensive price a t -shirt could be?

\$10.00

- b Write an inequality to represent the possible costs of t -shirts, in dollars.

Sample response: $x > 9.99$, where x represents the cost of t -shirts, in dollars

7. Priya looks at a container of cherries and says, "I think there are more than 55 cherries in the container."

Jada looks at the same container of cherries and says, "I think there are less than 125 cherries in the container."

- a Write an inequality to show Priya's statement, using c to represent the number of cherries.

$c > 55$ (or equivalent)

- b Write another inequality to show Jada's statement, also using c to represent the number of cherries.

$c < 125$ (or equivalent)

- c Can Priya and Jada both be correct? Explain your thinking and provide a possible number of cherries that supports your argument.

Yes; Sample response: Priya and Jada can both be correct because there are many numbers that are greater than 55, but less than 125. Some examples are 56, 100, 124.

8. Priya and Jada are each analyzing a container of blueberries. Study their statements. Did Priya and Jada each represent their statements with correct inequalities? Explain your thinking.

Priya

I think there are less than 150 blueberries in the container. The inequality that represents this is $b > 150$, where b represents the number of blueberries.

Jada

I think there are more than 65 blueberries in the container. The inequality that represents this is $b < 65$, where b represents the number of blueberries.

No; Sample response: Neither inequality statement is correct. Priya thinks there are less than 150 blueberries, so the correct inequality would be $b < 150$. Jada thinks there are more than 65 blueberries, so the correct inequality would be $b > 65$.

Name: _____ Date: _____ Period: _____

Additional Practice

7.07

- At a restaurant, a minimum of 2 people in each party need to be present in order to be seated at their table. Write an inequality that represents the possible number of people p that need to be present in order to be seated.
 $p \geq 2$ (or equivalent)
- Clare is younger than Bard. Bard is 13 years old. Write an inequality that compares Clare's age in years c to Bard's age.
 $c < 13$ (or equivalent)
- Diego started cooking dinner before 5:00 p.m. and finished cooking dinner at 6:00 p.m. Let d represent the number of hours Diego spent cooking dinner. Determine whether each statement is *definitely true*, *definitely not true*, or *possibly true*.

a $d < 1$ Definitely not true	b $d > 1$ Definitely true
c $d < 2$ Possibly true	d $d > 2$ Possibly true
e $d < 0.5$ Definitely not true	f $d > 0.5$ Definitely true
- Noah started a run at 6:30 a.m. and finished sometime after 9:00 a.m. Let r represent the number of hours Noah spent running. Determine whether each statement is *definitely true*, *definitely not true*, or *possibly true*.

a $r > 2$ Definitely true	b $r > 2.5$ Definitely true
c $r > 3$ Possibly true	d $r < 3$ Possibly true
e $r < 2.5$ Definitely not true	f $r < 2$ Definitely not true

Unit 7 Lesson 7

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Name: _____ Date: _____ Period: _____

- At the grocery store, all apples cost less than \$2.00 per pound.
 - What is the most expensive price one pound of apples could cost?
\$1.99
 - Write an inequality to represent the possible costs, in dollars, for any number of pounds of apples.
Sample response: $x < 2$, where x represents the cost, in dollars, for any number of pounds of apples
- At the clothing store, all t-shirts cost more than \$9.99.
 - What is the least expensive price a t-shirt could be?
\$10.00
 - Write an inequality to represent the possible costs of t-shirts, in dollars.
Sample response: $x > 9.99$, where x represents the cost of t-shirts, in dollars
- Priya looks at a container of cherries and says, "I think there are more than 55 cherries in the container." Jada looks at the same container of cherries and says, "I think there are less than 125 cherries in the container."
 - Write an inequality to show Priya's statement, using c to represent the number of cherries.
 $c > 55$ (or equivalent)
 - Write another inequality to show Jada's statement, also using c to represent the number of cherries.
 $c < 125$ (or equivalent)
 - Can Priya and Jada both be correct? Explain your thinking and provide a possible number of cherries that supports your argument.
Yes; Sample response: Priya and Jada can both be correct because there are many numbers that are greater than 55, but less than 125. Some examples are 56, 100, 124.
- Priya and Jada are each analyzing a container of blueberries. Study their statements. Did Priya and Jada each represent their statements with correct inequalities? Explain your thinking.

Priya
I think there are less than 150 blueberries in the container. The inequality that represents this is $b > 150$, where b represents the number of blueberries.

Jada
I think there are more than 65 blueberries in the container. The inequality that represents this is $b < 65$, where b represents the number of blueberries.

No; Sample response: Neither inequality statement is correct. Priya thinks there are less than 150 blueberries, so the correct inequality would be $b < 150$. Jada thinks there are more than 65 blueberries, so the correct inequality would be $b > 65$.

Unit 7 Lesson 7

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Additional Practice

Practice Problem Analysis

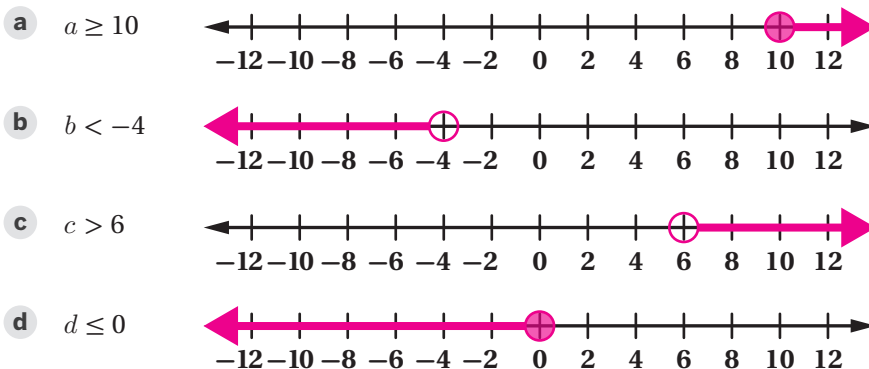
Problem	DOK	Standard(s)
1	1	6.EE.B.8
2	1	6.EE.B.8
3	2	6.EE.B.8
4	2	6.EE.B.8
5	2	6.EE.B.8
6	2	6.EE.B.8
7	3	6.EE.B.8
8	3	6.EE.B.8

Notes:

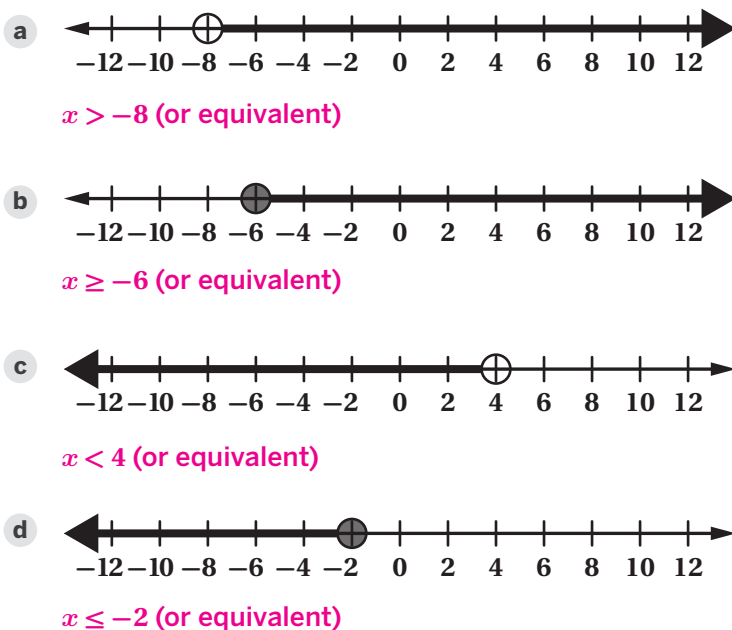
Additional Practice

7.08

1. Graph each inequality statement on the number line.



2. Use the variable x to write an inequality statement that represents each graph.



3. Consider the inequality $m < 4$.

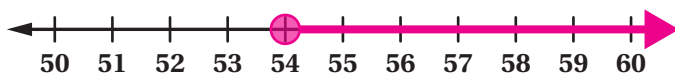
- a Which of these numbers are solutions to the inequality: 4, -3.1 , -1 , 4.6 , 1.5 , -8 , 5 ?
List *all* that apply.
 -3.1 , -1 , 1.5 , -8
- b Graph the inequality statement on the number line.
- 

4. A sign at an amusement park reads, “You must be 54 in. or taller to ride.”

- a Using the variable h , write an inequality that represents the information on the sign.

$$h \geq 54$$

- b Graph the inequality statement on the number line.



5. There are less than 12 people in line to ride the roller coaster.

- a Using the variable n , write an inequality that represents the number of people in line to ride the roller coaster.

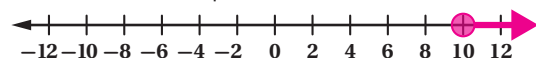
$$n < 12$$

- b Graph the inequality statement on the number line.



6. Represent each inequality scenario on the number line.

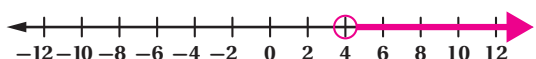
- a The amusement park sells at least 10 flavors of ice cream.



- b Andre spends less than \$6 at the arcade.



- c It costs more than 4 tokens for the ride.



7. Andre counts more than 3 hot dog stands at the fair. He said, “To graph this scenario on a number line, I would draw a closed circle at 3 with an arrow pointing to the right.” Did Andre correctly describe how to graph this scenario? Explain your thinking.

No; Sample response: Andre should have used an open circle at 3 because he saw more than 3 hot dog stands. The boundary value 3 is not included in the solution.

8. Priya and Jada each wrote a statement to describe a container of blueberries. Did Priya and Jada each correctly describe how to graph their inequality statements? Explain your thinking.

Priya

I think there are less than 150 blueberries in the container. To graph this inequality statement, I would draw an open circle at 150 with an arrow pointing to the right.

Jada

I think there are more than 65 blueberries in the container. To graph this inequality statement, I would draw a closed circle at 65 with an arrow pointing to the right.

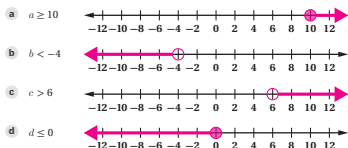
No; Sample response: Priya said there are less than 150 blueberries, so her arrow should point to the left, not to the right. Jada said there are more than 65 blueberries, which does not include 65, so she should use an open circle at 65.

Name: _____ Date: _____ Period: _____

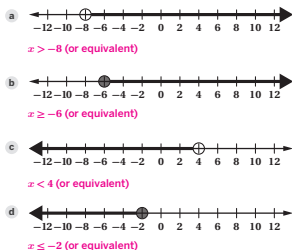
Additional Practice

7.08

1. Graph each inequality statement on the number line.



2. Use the variable x to write an inequality statement that represents each graph.



3. Consider the inequality $m < 4$.

- a Which of these numbers are solutions to the inequality: 4, -3.1, -1, 4.6, 1.5, -8, 5?
List all that apply.
-3.1, -1, 1.5, -8
- b Graph the inequality statement on the number line.

Unit 7 Lesson 8

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Name: _____ Date: _____ Period: _____

4. A sign at an amusement park reads, "You must be 54 in. or taller to ride."

- a Using the variable h , write an inequality that represents the information on the sign.
 $h \geq 54$

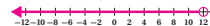
- b Graph the inequality statement on the number line.



5. There are less than 12 people in line to ride the roller coaster.

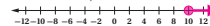
- a Using the variable n , write an inequality that represents the number of people in line to ride the roller coaster.
 $n < 12$

- b Graph the inequality statement on the number line.

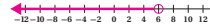


6. Represent each inequality scenario on the number line.

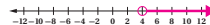
- a The amusement park sells at least 10 flavors of ice cream.



- b Andre spends less than \$6 at the arcade.



- c It costs more than 4 tokens for the ride.



7. Andre counts more than 3 hot dog stands at the fair. He said, "To graph this scenario on a number line, I would draw a closed circle at 3 with an arrow pointing to the right." Did Andre correctly describe how to graph this scenario? Explain your thinking.

No; Sample response: Andre should have used an open circle at 3 because he saw more than 3 hot dog stands. The boundary value 3 is not included in the solution.

8. Priya and Jada each wrote a statement to describe a container of blueberries. Did Priya and Jada each correctly describe how to graph their inequality statements? Explain your thinking.

Priya
I think there are less than 150 blueberries in the container. To graph this inequality statement, I would draw an open circle at 150 with an arrow pointing to the right.

Jada
I think there are more than 65 blueberries in the container. To graph this inequality statement, I would draw a closed circle at 65 with an arrow pointing to the right.

No; Sample response: Priya said there are less than 150 blueberries, so her arrow should point to the left, not to the right. Jada said there are more than 65 blueberries, which does not include 65, so she should use an open circle at 65.

Unit 7 Lesson 8

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	6.EE.B.8
2	1	6.EE.B.8
3	1	6.NS.C.7.B, 6.EE.B.8
4	2	6.EE.B.8
5	2	6.EE.B.8
6	2	6.NS.C.7.B, 6.EE.B.8
7	3	6.NS.C.7.B, 6.EE.B.8
8	3	6.NS.C.7.B, 6.EE.B.8

Notes:

Additional Practice

ACC7.3.18

1. For each scenario, circle the inequality that represents it.

- a The temperature is above 32°F.

$x > 32$

$x < 32$

$x \geq 32$

$x \leq 32$

- b Kiran has no more than \$20 to spend.

$x > 20$

$x < 20$

$x \geq 20$

$x \leq 20$

2. Consider the inequality $c \leq 9$. Select *all* the solutions to the inequality.

☒ A. 7

☒ B. 8

☒ C. 9

☐ D. 10

☐ E. 11

3. Write an inequality that represents each verbal description. Then write two values that will make the inequality true.

- a x is greater than 2.

$x > 2$. Any two values that satisfy $x > 2$ are considered correct.

- b b is less than or equal 4.5.

$b \leq 4.5$. Any two values that satisfy $b \leq 4.5$ are considered correct.

- c 5 is at least w .

$5 \geq w$ or $w \leq 5$. Any two values that satisfy $5 \geq w$ are considered correct.

- d d is no less than 16.

$d \geq 16$. Any two values that satisfy $d \geq 16$ are considered correct.

- e t is at most 8.

$t \leq 8$. Any two values that satisfy $t \leq 8$ are considered correct.

4. Write an inequality that represents each verbal description.

- a A student must have at least 5 hours of community service completed.

$x \geq 5$ (or equivalent)

- b Practice will be no more than 50 minutes.

$x \leq 50$ (or equivalent)

- c Children under the age of 3 are free.

$x < 3$ (or equivalent)

- d During hibernation, an arctic ground squirrel's body temperature never goes below -2.9°C .

$x \geq -2.9$ (or equivalent)

5. Consider the inequality $-2x < 14$.

- a List four values for x that would make this inequality true.

Responses vary. Values should be greater than -7 .

- b How are the solutions to the inequality $-2x < 14$ different from the solutions to the inequality $-2x \leq 14$?

Solutions to the inequality $-2x \leq 14$ include -7 .

6. Noah and Elena each wrote an inequality to represent the following situation.

A student wants to take, at minimum, \$15 to the store.

Noah's inequality: $x \geq 15$

Elena's inequality: $15 \leq x$

Who is correct? Circle one.

Noah

Elena

Both

Neither

Explain your thinking.

Explanations vary. The inequalities have the same solution.

7. In the first week of a canned food drive 480 cans were collected. In the last week of the canned food drive, at least twice that amount was collected. Write an inequality that represents the amount of cans that were collected in the last week of the canned food drive.

$x \geq 960$ (or equivalent)

8. The number line shows a solution to an inequality. Write a real-world scenario that the inequality could represent.



Responses vary. A student has at least 10 pencils in their backpack at all times.

Name: _____ Date: _____ Period: _____

Additional Practice

ACC7.3.18

1. For each scenario, circle the inequality that represents it.

a. The temperature is above 32°F.

☒ $x > 32$ ☐ $x < 32$ ☐ $x \geq 32$ ☐ $x \leq 32$

b. Kiran has no more than \$20 to spend.

☐ $x > 20$ ☐ $x < 20$ ☐ $x \geq 20$ ☒ $x \leq 20$

2. Consider the inequality $c \leq 9$. Select *all* the solutions to the inequality.

☒ A. 7 ☒ B. 8 ☒ C. 9 ☐ D. 10 ☐ E. 11

3. Write an inequality that represents each verbal description. Then write two values that will make the inequality true.

a. x is greater than 2.

$x > 2$. Any two values that satisfy $x > 2$ are considered correct.

b. b is less than or equal 4.5.

$b \leq 4.5$. Any two values that satisfy $b \leq 4.5$ are considered correct.

c. 5 is at least w .

$5 \geq w$ or $w \leq 5$. Any two values that satisfy $5 \geq w$ are considered correct.

d. d is no less than 16.

$d \geq 16$. Any two values that satisfy $d \geq 16$ are considered correct.

e. t is at most 8.

$t \leq 8$. Any two values that satisfy $t \leq 8$ are considered correct.

4. Write an inequality that represents each verbal description.

a. A student must have at least 5 hours of community service completed.

$x \geq 5$ (or equivalent)

b. Practice will be no more than 50 minutes.

$x \leq 50$ (or equivalent)

c. Children under the age of 3 are free.

$x < 3$ (or equivalent)

d. During hibernation, an arctic ground squirrel's body temperature never goes below -2.9°C .

$x \geq -2.9$ (or equivalent)

Unit 3 Lesson 18

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Name: _____ Date: _____ Period: _____

5. Consider the inequality $-2x < 14$.

a. List four values for x that would make this inequality true.

Responses vary. Values should be greater than -7 .

b. How are the solutions to the inequality $-2x < 14$ different from the solutions to the inequality $-2x \leq 14$?

Solutions to the inequality $-2x \leq 14$ include -7 .

6. Noah and Elena each wrote an inequality to represent the following situation.

A student wants to take, at minimum, \$15 to the store.

Noah's inequality: $x \geq 15$

Elena's inequality: $15 \leq x$

Who is correct? Circle one.

Noah Elena ☒ Both Neither

Explain your thinking.

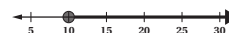
Explanations vary. The inequalities have the same solution.

7. In the first week of a canned food drive 480 cans were collected. In the last week of the canned food drive, at least twice that amount was collected. Write an inequality that represents the amount of cans that were collected in the last week of the canned food drive.

$x \geq 960$ (or equivalent)

8. The number line shows a solution to an inequality. Write a real-world scenario that the inequality could represent.

Responses vary. A student has at least 10 pencils in their backpack at all times.



Unit 3 Lesson 18

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	7.EE.B.4
2	1	7.EE.B.4
3	2	7.EE.B.4
4	2	7.EE.B.4
5	2	7.EE.B.4
6	2	7.EE.B.4
7	3	7.EE.B.4
8	3	7.EE.B.4

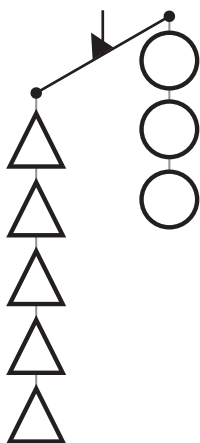
Notes:

Additional Practice

ACC7.3.19

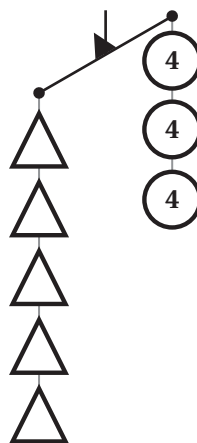
Problems 1–2: Here are two unbalanced hangers. Write an inequality to represent the relationship between the weights on each hanger. Use t to represent the weight of the triangle in grams. Use c to represent the weight of the circle in grams.

1.



$5t > 3c$ (or equivalent)

2.



$t > 2\frac{2}{5}$ (or equivalent)

3. Select *all* values of x that make the inequality $x - 7 \geq -3$ true.

☐ A. 3.9

☐ B. 0

☒ C. 4

☒ D. 4.1

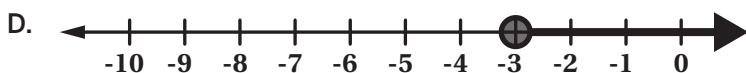
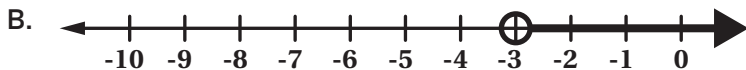
☒ E. 5

☐ F. -1

☐ G. -4

☒ H. 8

4. Which number line shows the solution to $-3x > 9$?



5. Melanie solves the inequality $-5x \leq -40$. She first solves the equation $-5x = -40$ and determines $x = 8$. What is the solution to the inequality?

A. $x \leq 8$

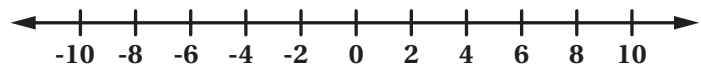
B. $x \geq 8$

C. $x < 8$

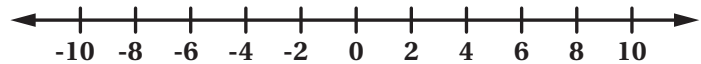
D. $x > 8$

Problems 6–7: Use the number line diagrams to graph possible solutions for each inequality.

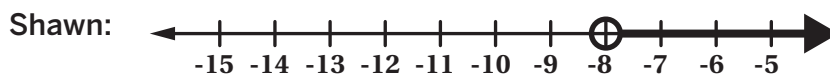
6. $-3x < 18$



7. $\frac{2}{3}x \leq \frac{14}{3}$



8. Shawn solved the inequality $-\frac{1}{2}x > 4$. Shawn's solution is shown.



Is Shawn correct? Explain your thinking.

Shawn is incorrect. Explanations vary. Shawn is correct that the solution to the equation $-\frac{1}{2}x = 4$ is $x = -8$. However, when testing the value $x = 2$ in the inequality $-\frac{1}{2}x > 4$, he would get a false statement: $-\frac{1}{2}(2) > 4 \rightarrow -1 > 4$. So, the solution would be values *less than* -8 .

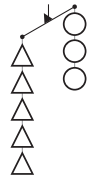
Name: _____ Date: _____ Period: _____

Additional Practice

ACC7.3.19

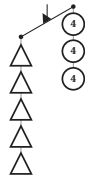
Problems 1–2: Here are two unbalanced hangers. Write an inequality to represent the relationship between the weights on each hanger. Use t to represent the weight of the triangle in grams. Use c to represent the weight of the circle in grams.

1.



$5t > 3c$ (or equivalent)

2.

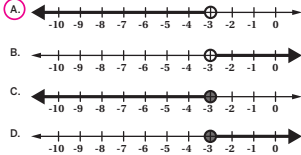


$t > \frac{2}{5}c$ (or equivalent)

3. Select all values of x that make the inequality $x - 7 \geq -3$ true.

- ☐ A. 3.9 ☐ B. 0
☒ C. 4 ☒ D. 4.1
☒ E. 5 ☐ F. -1
☐ G. -4 ☒ H. 8

4. Which number line shows the solution to $-3x > 9$?



Unit 3 Lesson 19

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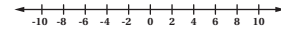
Name: _____ Date: _____ Period: _____

5. Melanie solves the inequality $-5x \leq -40$. She first solves the equation $-5x = -40$ and determines $x = 8$. What is the solution to the inequality?

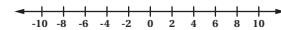
- ☐ A. $x \leq 8$ ☒ B. $x \geq 8$
☐ C. $x < 8$ ☐ D. $x > 8$

Problems 6–7: Use the number line diagrams to graph possible solutions for each inequality.

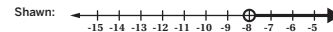
6. $-3x \leq 18$



7. $\frac{2}{3}x \leq \frac{14}{3}$



8. Shawn solved the inequality $-\frac{1}{2}x > 4$. Shawn's solution is shown.



Is Shawn correct? Explain your thinking.

Shawn is incorrect. *Explanations vary.* Shawn is correct that the solution to the equation $-\frac{1}{2}x = 4$ is $x = -8$. However, when testing the value $x = 2$ in the inequality $-\frac{1}{2}x > 4$, he would get a false statement: $-\frac{1}{2}(2) > 4 \rightarrow -1 > 4$. So, the solution would be values less than -8 .

Unit 3 Lesson 19

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	7.EE.B.4
2	2	7.EE.B.4
3	1	7.EE.B.4
4	1	7.EE.B.4.B
5	1	7.EE.B.4.B
6	1	7.EE.B.4.B
7	1	7.EE.B.4.B
8	2	7.EE.B.4.B

Notes:

Additional Practice

ACC7.3.20

1. Priya has \$30 to spend at the school festival. Admission is \$4 and each ride ticket is \$2. Which inequality represents the greatest number of ride tickets she can buy?

A. $2n + 4 < 30$

B. $2n + 4 > 30$

C. $2n + 4 \leq 30$

D. $2n + 4 \geq 30$

2. It is currently 8 degrees outside, and the temperature will drop 2 degrees every hour. When the temperature falls below 0 degrees, it can be represented by the inequality $8 - 2h < 0$, where h represents the number hours the temperature has been dropping. Does $h < 4$ or $h > 4$ represent the solution of the inequality?

$h > 4$

3. The 20 members of the photography club are trying to raise at least \$1,400 for new photography equipment. They have already raised \$540.

- a Let m represent the amount of money each member must raise, on average, to meet their goal. Write an expression for the total amount of money going to be raised.

$540 + 20m$

- b Write an equation that represents the club raising all the money.

$540 + 20m = 1400$

- c Solve the equation. What does the solution mean in context of the scenario?

$m = 43$. Each member needs to raise \$43.

- d Write an inequality representing the amount of money each member must raise, on average, to meet or exceed their goal.

$540 + 20m \geq 1400$

- e Write an inequality showing the possible average amount of money each club member needs to raise to meet or exceed their goal.

$m \geq 43$

4. A baker has 84 pounds of flour. After using p pounds of flour each month for 6 months, the amount of flour the baker has left is $84 - 6p$.

a Explain what the equation $84 - 6p = 76$ represents.

It represents when the baker will have 76 pounds of flour.

b What value of p makes the equation true?

$p = 1\frac{1}{3}$ (or equivalent)

c Explain what the inequality $84 - 6p \leq 76$ represents.

It represents when the baker will have 76 pounds of flour or less.

d What values of p make the inequality true?

Responses must be greater than or equal to $1\frac{1}{3}$.

5. Noah spent \$40 on supplies for making 25 birdhouses. Noah wants to make a profit of more than \$300.

a Write an inequality to determine the price p Noah should charge per birdhouse if he wants to meet his goal.

$25p - 40 > 300$ (or equivalent)

b What values of p make the inequality true?

$p > 13.60$ (or equivalent)

6. Elena scored 95, 91, and 90 on 3 tests. She wants her average test score for 5 tests to be at least 92.

a Elena wants to know what she needs to get on her next two tests to meet her goal. Write an inequality where x represents what Elena will need to score on the next two tests.

$276 + 2x \geq 460$ (or equivalent)

b What values make the inequality true?

$x \geq 92$

Name: _____ Date: _____ Period: _____

Additional Practice

ACC7.3.20

- Priya has \$30 to spend at the school festival. Admission is \$4 and each ride ticket is \$2. Which inequality represents the greatest number of ride tickets she can buy?
A. $2n + 4 < 30$ B. $2n + 4 > 30$ **C. $2n + 4 \leq 30$** D. $2n + 4 \geq 30$
- It is currently 8 degrees outside, and the temperature will drop 2 degrees every hour. When the temperature falls below 0 degrees, it can be represented by the inequality $8 - 2h < 0$, where h represents the number hours the temperature has been dropping. Does $h < 4$ or $h > 4$ represent the solution of the inequality?
 $h > 4$
- The 20 members of the photography club are trying to raise at least \$1,400 for new photography equipment. They have already raised \$540.
 - Let m represent the amount of money each member must raise, on average, to meet their goal. Write an expression for the total amount of money going to be raised.
 $540 + 20m$
 - Write an equation that represents the club raising all the money.
 $540 + 20m = 1400$
 - Solve the equation. What does the solution mean in context of the scenario?
 $m = 43$. Each member needs to raise \$43.
 - Write an inequality representing the amount of money each member must raise, on average, to meet or exceed their goal.
 $540 + 20m \geq 1400$
 - Write an inequality showing the possible average amount of money each club member needs to raise to meet or exceed their goal.
 $m \geq 43$

Unit 3 Lesson 20

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Name: _____ Date: _____ Period: _____

- A baker has 84 pounds of flour. After using p pounds of flour each month for 6 months, the amount of flour the baker has left is $84 - 6p$.
 - Explain what the equation $84 - 6p = 76$ represents.
It represents when the baker will have 76 pounds of flour.
 - What value of p makes the equation true?
 $p = 1\frac{1}{3}$ (or equivalent)
 - Explain what the inequality $84 - 6p \leq 76$ represents.
It represents when the baker will have 76 pounds of flour or less.
 - What values of p make the inequality true?
Responses must be greater than or equal to $1\frac{1}{3}$.
- Noah spent \$40 on supplies for making 25 birdhouses. Noah wants to make a profit of more than \$300.
 - Write an inequality to determine the price p Noah should charge per birdhouse if he wants to meet his goal.
 $25p - 40 > 300$ (or equivalent)
 - What values of p make the inequality true?
 $p > 13.60$ (or equivalent)
- Elena scored 95, 91, and 90 on 3 tests. She wants her average test score for 5 tests to be at least 92.
 - Elena wants to know what she needs to get on her next two tests to meet her goal. Write an inequality where x represents what Elena will need to score on the next two tests.
 $276 + 2x \geq 460$ (or equivalent)
 - What values make the inequality true?
 $x \geq 92$

Unit 3 Lesson 20

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	7.EE.B.4.B
2	1	7.EE.B.4.B
3	2	7.EE.B.4
4	2	7.EE.B.4.B
5	2	7.EE.B.4
6	3	7.EE.B.4

Notes:

Additional Practice

ACC7.3.21

1. Select *all* the values of x that make the inequality $-x + 6 < 8$ true.

☒ A. 1.9

☒ B. 2

☒ C. 2.01

☐ D. -2.01

☐ E. -2

☒ F. -1.9

☒ G. 15

☐ H. -15

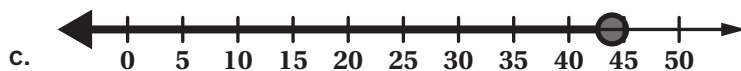
2. To ride on the Tall Mountain rollercoaster, guests must be at least 44 inches tall. Match each situation or graph with an inequality.

a. Kira is x inches tall and cannot ride the rollercoaster.

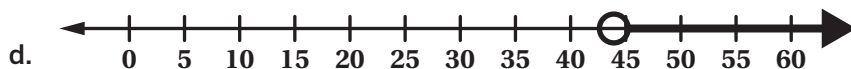
d $x > 44$

b. Jake is x inches tall and can ride the rollercoaster.

c $x \leq 44$



a $x < 44$



b $x \geq 44$

Problems 3–4: Here is the inequality $60 - 2x \geq -10$.

3. Select *all* the values of x that make the inequality $60 - 2x \geq -10$ true.

☒ A. 0

☒ B. 35

☒ C. -35

☐ D. 35.1

☒ E. 34.9

4. In order to solve the inequality $60 - 2x \geq -10$, Soorya solved the equation $60 - 2x = -10$ and got $x = 35$. What is the solution to the inequality? Show or explain your thinking.

$x \leq 35$. Explanations vary. I substituted the value $x = 1$ into the inequality $60 - 2x \geq -10$, and I got $60 - 2 \geq -10$ or $58 \geq -10$, which is a true statement. Therefore, the solution is all values less than or equal to 35.

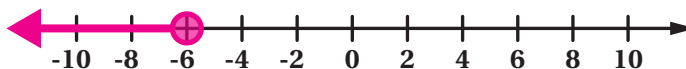
5. Jasmine is solving the inequality $12x < -48$. She solves the equation $12x = -48$ to determine $x = -4$. What is the solution to the inequality?

- A. $x < -4$
 B. $x > -4$
 C. $x \leq -4$
 D. $x \geq -4$

Problems 6–7: Complete the table to determine the solutions to each inequality. Write the solutions as an inequality and graph them on the number line.

6. $-5x \geq 30$

x	-9	-8	-7	-6	-5	-4	-3
$-5x$	45	40	35	30	25	20	15



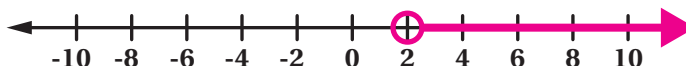
7. $\frac{4}{5}x > -12$

x	-20	-15	-10	-5	0	5	10
$\frac{4}{5}x$	-16	-12	-8	-4	0	4	8



8. Solve the inequality $2x + 8 > 12$ and graph the solution on the number line. Show your thinking.

$x > 2$



Name: _____ Date: _____ Period: _____

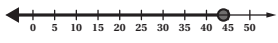
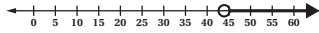
Additional Practice

ACC7.3.21

1. Select all the values of x that make the inequality $-x + 6 < 8$ true.

- ☒ A. 1.9 ☒ B. 2
☒ C. -2.01 ☐ D. -2.01
☐ E. -2 ☒ F. -1.9
☒ G. 15 ☐ H. -15

2. To ride on the Tall Mountain rollercoaster, guests must be at least 44 inches tall. Match each situation or graph with an inequality.

- a. Kira is x inches tall and cannot ride the rollercoaster. d. $x > 44$
b. Jake is x inches tall and can ride the rollercoaster. c. $x \leq 44$
c.  a. $x < 44$
d.  b. $x \geq 44$

Problems 3–4: Here is the inequality $60 - 2x \geq -10$.

3. Select all the values of x that make the inequality $60 - 2x \geq -10$ true.

- ☒ A. 0 ☒ B. 35 ☒ C. -35
☐ D. 35.1 ☒ E. 34.9

4. In order to solve the inequality $60 - 2x \geq -10$, Soorya solved the equation $60 - 2x = -10$ and got $x = 35$. What is the solution to the inequality? Show or explain your thinking.

$x \leq 35$. *Explanations vary. I substituted the value $x = 1$ into the inequality $60 - 2x \geq -10$, and I got $60 - 2(1) \geq -10$ or $58 \geq -10$, which is a true statement. Therefore, the solution is all values less than or equal to 35.*

Unit 3 Lesson 21

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Name: _____ Date: _____ Period: _____

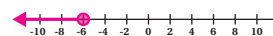
5. Jasmine is solving the inequality $12x < -48$. She solves the equation $12x = -48$ to determine $x = -4$. What is the solution to the inequality?

- ☒ A. $x < -4$
☐ B. $x > -4$
☐ C. $x \leq -4$
☐ D. $x \geq -4$

Problems 6–7: Complete the table to determine the solutions to each inequality. Write the solutions as an inequality and graph them on the number line.

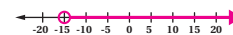
6. $-5x \geq 30$

x	-9	-8	-7	-6	-5	-4	-3
$-5x$	45	40	35	30	25	20	15



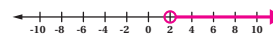
7. $\frac{4}{5}x > -12$

x	-20	-15	-10	-5	0	5	10
$\frac{4}{5}x$	-16	-12	-8	-4	0	4	8



8. Solve the inequality $2x + 8 > 12$ and graph the solution on the number line. Show your thinking.

$x > 2$



Unit 3 Lesson 21

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	7.EE.B.3
2	2	7.EE.B.4.B
3	1	7.EE.B.3
4	1	7.EE.B.4.B
5	1	7.EE.B.4.B
6	1	7.EE.B.3, 7.EE.B.4.B
7	1	7.EE.B.3, 7.EE.B.4.B
8	1	7.EE.B.3, 7.EE.B.4.B

Notes:

Additional Practice

ACC7.3.22

Problems 1–3: A bakery sold $\frac{2}{3}$ of the loaves of bread on display and brought out another 10. The bakery likes to keep at least 25 loaves of bread on display. The manager wrote the inequality $\frac{1}{3}x + 10 \geq 25$ to describe the situation.

1. Explain what $\frac{1}{3}x$ means in the manager's inequality.

It represents the number of loaves of bread that remained in the display. After $\frac{2}{3}$ of the bread on display was sold, $\frac{1}{3}$ of the loaves of bread remain on display.

2. Solve the inequality.

$$x \geq 45$$

3. What do the solutions to the inequality mean in this situation?

Initially, the bakery had at least 45 loaves of bread on display.

Problems 4–6: A teacher has up to \$150 to spend on classroom supplies. He plans to purchase 12 packs of markers and plans to spend \$45.25 on posters.

4. Write an inequality to represent the situation.

$$12x + 45.25 \leq 150 \text{ (or equivalent)}$$

5. Solve the inequality.

$$x \leq 8.73 \text{ (or equivalent)}$$

6. What do the solutions to your inequality mean in this situation?

The teacher can spend no more than \$8.73 for each pack of markers.

Problems 7–8: Write an inequality to represent each situation.

- 7.** A library has one large meeting room that fits 20 people and many smaller rooms that hold 6 people. There are enough meeting rooms to hold at most 50 people. Write an inequality whose solution is the possible number of 6-person meeting rooms in the library.

$20 + 6x \leq 50$ (or equivalent)

- 8.** There are 7 recycling bins in a school cafeteria. 6 recycling bins are the same size, and the seventh holds 20 liters of recycling. Students for Environmental Action are hoping that the recycling bins can hold at least 50 liters of recycling. Write an inequality whose solution is the possible size of each of the 6 recycling bins.

$20 + 6x \geq 50$ (or equivalent)

Problems 9–11: Here is the inequality $2x + 8 \leq 35$.

- 9.** Write a situation to represent the inequality. Be sure to describe what the variable represents.

Responses vary. I am going to the movies and have at most \$35 to spend. I am going to purchase 2 tickets, one for me and one for a friend, and buy an \$8 popcorn for us to share. Let x represent the cost of the movie ticket.

- 10.** Solve the inequality.

$x \leq 13.50$ (or equivalent)

- 11.** What do the solutions to the inequality mean in your situation?

Responses vary. I can spend at most \$13.50 on each movie ticket.

Name: _____ Date: _____ Period: _____

Additional Practice

ACC7.3.22

Problems 1–3: A bakery sold $\frac{2}{3}$ of the loaves of bread on display and brought out another 10. The bakery likes to keep at least 25 loaves of bread on display. The manager wrote the inequality $\frac{1}{3}x + 10 \geq 25$ to describe the situation.

1. Explain what $\frac{1}{3}x$ means in the manager's inequality.
It represents the number of loaves of bread that remained in the display. After $\frac{2}{3}$ of the bread on display was sold, $\frac{1}{3}$ of the loaves of bread remain on display.

2. Solve the inequality.
 $x \geq 45$

3. What do the solutions to the inequality mean in this situation?
Initially, the bakery had at least 45 loaves of bread on display.

Problems 4–6: A teacher has up to \$150 to spend on classroom supplies. He plans to purchase 12 packs of markers and plans to spend \$45.25 on posters.

4. Write an inequality to represent the situation.
 $12x + 45.25 \leq 150$ (or equivalent)
5. Solve the inequality.
 $x \leq 8.73$ (or equivalent)
6. What do the solutions to your inequality mean in this situation?
The teacher can spend no more than \$8.73 for each pack of markers.

Unit 3 Lesson 22

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Name: _____ Date: _____ Period: _____

Problems 7–8: Write an inequality to represent each situation.

7. A library has one large meeting room that fits 20 people and many smaller rooms that hold 6 people. There are enough meeting rooms to hold at most 50 people. Write an inequality whose solution is the possible number of 6-person meeting rooms in the library.
 $20 + 6x \leq 50$ (or equivalent)

8. There are 7 recycling bins in a school cafeteria. 6 recycling bins are the same size, and the seventh holds 20 liters of recycling. Students for Environmental Action are hoping that the recycling bins can hold at least 50 liters of recycling. Write an inequality whose solution is the possible size of each of the 6 recycling bins.
 $20 + 6x \geq 50$ (or equivalent)

Problems 9–11: Here is the inequality $2x + 8 \leq 35$.

9. Write a situation to represent the inequality. Be sure to describe what the variable represents.

Responses vary. I am going to the movies and have at most \$35 to spend. I am going to purchase 2 tickets, one for me and one for a friend, and buy an \$8 popcorn for us to share. Let x represent the cost of the movie ticket.

10. Solve the inequality.
 $x \leq 13.50$ (or equivalent)

11. What do the solutions to the inequality mean in your situation?
Responses vary. I can spend at most \$13.50 on each movie ticket.

Unit 3 Lesson 22

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	7.EE.B.4
2	2	7.EE.B.4.B
3	2	7.EE.B.4.B
4	2	7.EE.B.4.B
5	2	7.EE.B.4.B
6	2	7.EE.B.4.B
7	2	7.EE.B.4.B
8	2	7.EE.B.4.B
9	2	7.EE.B.4
10	2	7.EE.B.4.B
11	2	7.EE.B.4.B

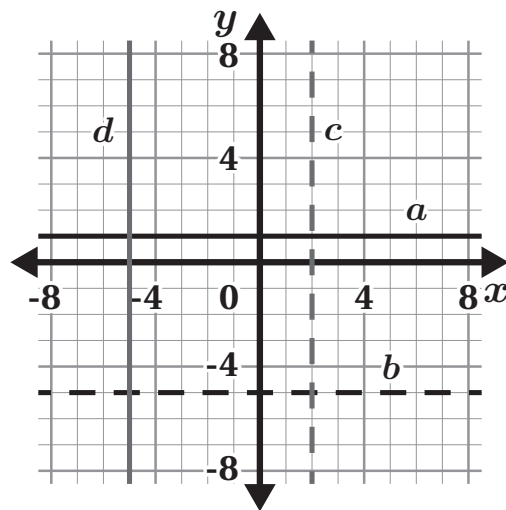
Notes:

Additional Practice

ACC7.4.10

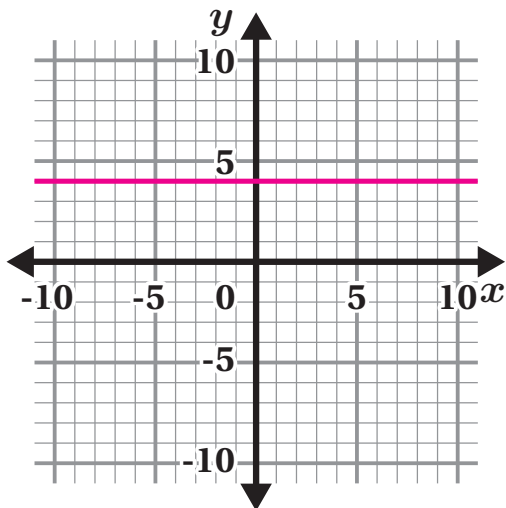
Problems 1–4: Here are four lines on a coordinate plane. Write an equation for each line.

1. Equation for a : $y = 1$
2. Equation for b : $y = -5$
3. Equation for c : $x = 2$
4. Equation for d : $x = -5$



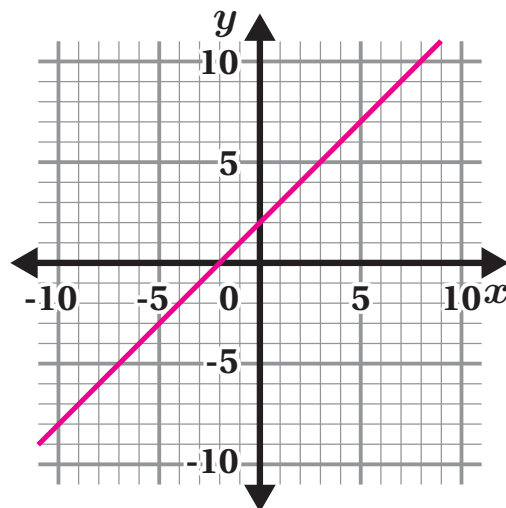
5. Draw a line with a slope of 0 and a y -intercept of 4. Then write an equation for the line.

$y = 4$



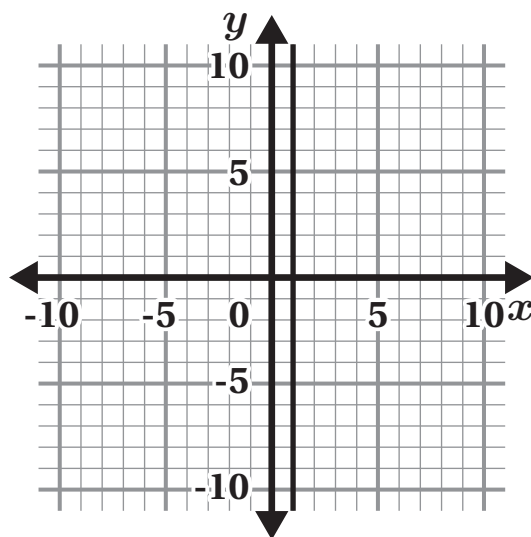
6. Draw a line with a slope of 1 and a y -intercept of 2. Then write an equation for the line.

$y = x + 2$ (or equivalent)



7. This graph represents a linear relationship.
Circle the equation it matches.

- A. $y = 1$
- B. $y = -1$
- C. $x = 1$**
- D. $x = -1$



8. Write an equation for the line that passes through the points (2, 7) and (5, 7).

$y = 7$

9. Write an equation for the line that passes through the points (9, -5) and (9, 13).

$x = 9$

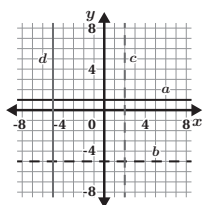
Name: _____ Date: _____ Period: _____

Additional Practice

ACC7.4.10

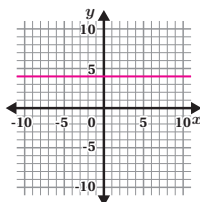
Problems 1–4: Here are four lines on a coordinate plane. Write an equation for each line.

1. Equation for a : $y = 1$
2. Equation for b : $y = -5$
3. Equation for c : $x = 2$
4. Equation for d : $x = -5$



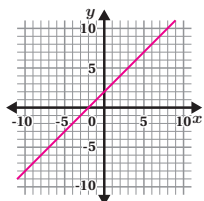
5. Draw a line with a slope of 0 and a y -intercept of 4. Then write an equation for the line.

$y = 4$



6. Draw a line with a slope of 1 and a y -intercept of 2. Then write an equation for the line.

$y = x + 2$ (or equivalent)



Unit 4 Lesson 10

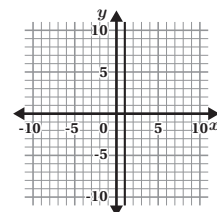
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7. This graph represents a linear relationship. Circle the equation it matches.

- A. $y = 1$
- B. $y = -1$
- C. $x = 1$**
- D. $x = -1$



8. Write an equation for the line that passes through the points (2, 7) and (5, 7).

$y = 7$

9. Write an equation for the line that passes through the points (9, -5) and (9, 13).

$x = 9$

Unit 4 Lesson 10

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Additional Practice

Practice Problem Item Analysis

Problem(s)	DOK	Standard(s)
1–4	1	8.EE.B
5–6	1	8.EE.B
7	1	8.EE.B, 8.F.B.4
8–9	2	8.EE.B

Notes:

Additional Practice

ACC7.4.11

1. Which equation represents the line that passes through points $(-8, 12)$ and $(4, 6)$?

A. $y = \frac{1}{2}x + 8$

B. $y = -\frac{1}{2}x + 8$

C. $y = \frac{1}{2}x - 2$

D. $y = -\frac{1}{2}x - 2$

2. Lin is writing an equation of the line that passes the point $(4, 24)$ and has a slope of 3. Her unfinished work is shown:

Lin's work:

$$\begin{aligned} y &= 3x + b & m &= 3 \\ 24 &= 3(4) + b \end{aligned}$$

Finish Lin's work by solving the equation for b and writing the final equation for the line in the form $y = mx + b$. Show your thinking.

Work varies.

$$24 = 3(4) + b$$

$$24 = 12 + b$$

$$12 = b$$

Equation: $y = 3x + 12$

3. Write the equation of the line that passes through each pair of points. Show or explain your thinking.

a $(4, 18)$ and $(8, 34)$

Responses vary.

$$\text{Slope: } \frac{34 - 18}{8 - 4} = \frac{16}{4} = 4$$

$$18 = 4(4) + b$$

$$18 = 16 + b$$

$$2 = b$$

Equation: $y = 4x + 2$

b $(-4, 14)$ and $(2, 2)$

Responses vary.

$$\text{Slope: } \frac{2 - 14}{2 - (-4)} = \frac{-12}{6} = -2$$

$$14 = -2(-4) + b$$

$$14 = 8 + b$$

$$6 = b$$

Equation: $y = -2x + 6$

4. Diego is finding the slope of the line that passes through the points $(10, 8)$ and $(14, 10)$. Determine any mistakes and explain how you would fix them.

Diego's work:

$$\text{Slope} = \frac{10 - 8}{10 - 14} = \frac{2}{-4} = -\frac{1}{2}$$

Responses vary. Diego did not correctly subtract the x -coordinates for the two points in the same order as the y -coordinates. The slope should be $\frac{10 - 8}{14 - 10} = \frac{2}{4} = \frac{1}{2}$.

5. A cooler of water is draining. After 4 minutes, there are 16 quarts remaining. After 6 minutes, there are 8 quarts remaining. Write an equation for the amount of water remaining in the cooler, y , after x minutes. Show or explain your thinking.

$$y = -4x + 32. \text{ Explanations vary.}$$

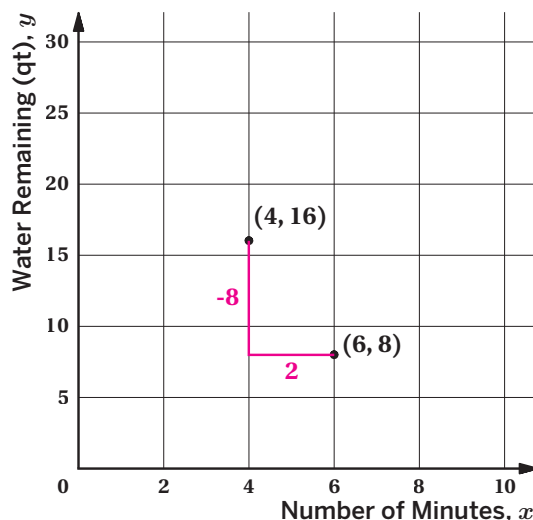
$$\text{Slope: } -4$$

$$8 = -4(6) + b$$

$$8 = -24 + b$$

$$32 = b$$

$$\text{Equation: } y = -4x + 32$$



6. The table gives the x - and y -coordinates of points on a line. Without graphing, write an equation for the line that passes through the points. Show or explain your thinking.

x	-4	4	8
y	14	2	-4

$$y = -\frac{3}{2}x + 8 \text{ Explanations vary.}$$

$$\text{Slope: } -\frac{3}{2}$$

$$2 = -\frac{3}{2}(4) + b$$

$$2 = -6 + b$$

$$8 = b$$

$$\text{Equation: } y = -\frac{3}{2}x + 8$$

7. Consider line m with the two labeled points as shown. Kiran missed today's lesson and does not understand how he can write the equation of the line without seeing the vertical intercept. Help Kiran understand how to write the equation for the line in the form $y = mx + b$.

Responses vary. First, calculate the slope of the line by determining the ratio of the difference in the y -coordinates to the difference in the x -coordinates:

$$\frac{-4 - 7}{4 - 3} = -\frac{11}{1} = -11. \text{ The slope is } -11.$$

Next, substitute the slope and the coordinates of one of the points into the equation $y = mx + b$. Then solve for b .

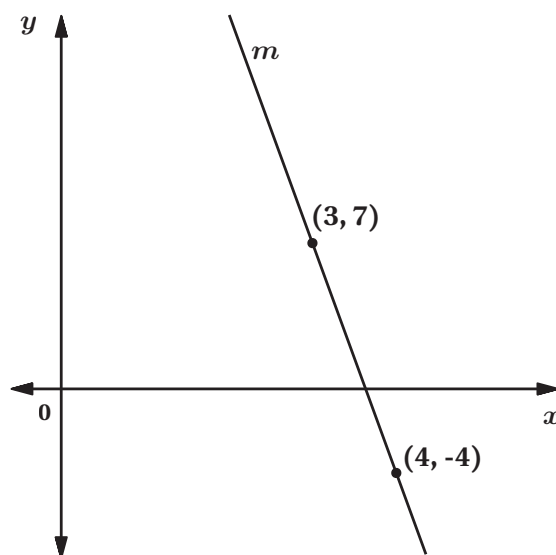
$$7 = -11(3) + b$$

$$7 = -33 + b$$

$$40 = b$$

Finally, write the equation in the form $y = mx + b$ using the slope and the y -intercept:

$$y = -11x + 40.$$



Name: _____ Date: _____ Period: _____

Additional Practice

ACC7.4.11

1. Which equation represents the line that passes through points $(-8, 12)$ and $(4, 6)$?

A. $y = \frac{1}{2}x + 8$
 B. $y = -\frac{1}{2}x + 8$
 C. $y = \frac{1}{2}x - 2$
 D. $y = -\frac{1}{2}x - 2$

2. Lin is writing an equation of the line that passes the point $(4, 24)$ and has a slope of 3. Her unfinished work is shown:

Lin's work:
 $y = 3x + b$
 $24 = 3(4) + b$
 $m = 3$

Finish Lin's work by solving the equation for b and writing the final equation for the line in the form $y = mx + b$. Show your thinking.

Work varies.
 $24 = 3(4) + b$
 $24 = 12 + b$
 $12 = b$

Equation: $y = 3x + 12$

3. Write the equation of the line that passes through each pair of points. Show or explain your thinking.

a. $(4, 18)$ and $(8, 34)$

Responses vary.
 Slope: $\frac{34 - 18}{8 - 4} = \frac{16}{4} = 4$
 $18 = 4(4) + b$
 $18 = 16 + b$
 $2 = b$

Equation: $y = 4x + 2$

b. $(-4, 14)$ and $(2, 2)$

Responses vary.
 Slope: $\frac{2 - 14}{2 - (-4)} = \frac{-12}{6} = -2$
 $14 = -2(-4) + b$
 $14 = 8 + b$
 $6 = b$

Equation: $y = -2x + 6$

4. Diego is finding the slope of the line that passes through the points $(10, 8)$ and $(14, 10)$. Determine any mistakes and explain how you would fix them.

Diego's work:
 Slope = $\frac{10 - 8}{10 - 14} = \frac{2}{-4} = -\frac{1}{2}$

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Unit 4 Lesson 11

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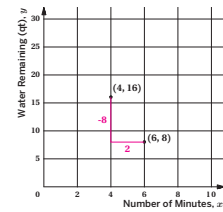
Name: _____ Date: _____ Period: _____

5. A cooler of water is draining. After 4 minutes, there are 16 quarts remaining. After 6 minutes, there are 8 quarts remaining. Write an equation for the amount of water remaining in the cooler, y , after x minutes. Show or explain your thinking.

$y = -4x + 32$. Explanations vary.

Slope: -4
 $8 = -4(6) + b$
 $8 = -24 + b$
 $32 = b$

Equation: $y = -4x + 32$



6. The table gives the x - and y -coordinates of points on a line. Without graphing, write an equation for the line that passes through the points. Show or explain your thinking.

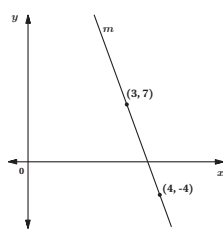
x	-4	4	8
y	14	2	-4

$y = -\frac{3}{2}x + 8$. Explanations vary.

Slope: $-\frac{3}{2}$
 $2 = -\frac{3}{2}(4) + b$
 $2 = -6 + b$
 $8 = b$

Equation: $y = -\frac{3}{2}x + 8$

7. Consider line m with the two labeled points as shown. Kiran missed today's lesson and does not understand how he can write the equation of the line without seeing the vertical intercept. Help Kiran understand how to write the equation for the line in the form $y = mx + b$.



Responses vary. First, calculate the slope of the line by determining the ratio of the difference in the y -coordinates to the difference in the x -coordinates:

$\frac{-4 - 7}{4 - 3} = \frac{-11}{1} = -11$. The slope is -11 .

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$7 = -11(3) + b$
 $7 = -33 + b$
 $40 = b$

Finally, write the equation in the form $y = mx + b$ using the slope and the y -intercept:
 $y = -11x + 40$.

Unit 4 Lesson 11

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.EE.B.6
2	1	8.EE.B.6
3	2	8.EE.B.6
4	2	8.EE.B.6
5	2	8.EE.B.6
6	2	8.EE.B.6
7	3	8.EE.B.6

Notes: